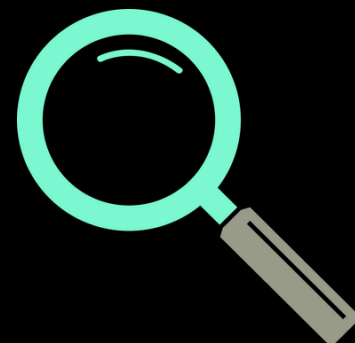




Intro to Digital Forensics

UofTCTF Seminars 2026



Announcements

- 2 roles we are hiring for:
 - VP finance
 - VP technology
 - view roles on sop + email us if you're interested
- No workshop during finals season, we will resume in May
- If you want to join our CCNA study group send me a DM

Outline

- File Forensics
- Steganography
- Network Forensics
- Logs & System Artifacts
- Memory Forensics

Forensics

Forensics is the process of collecting artifacts from digital traces like:

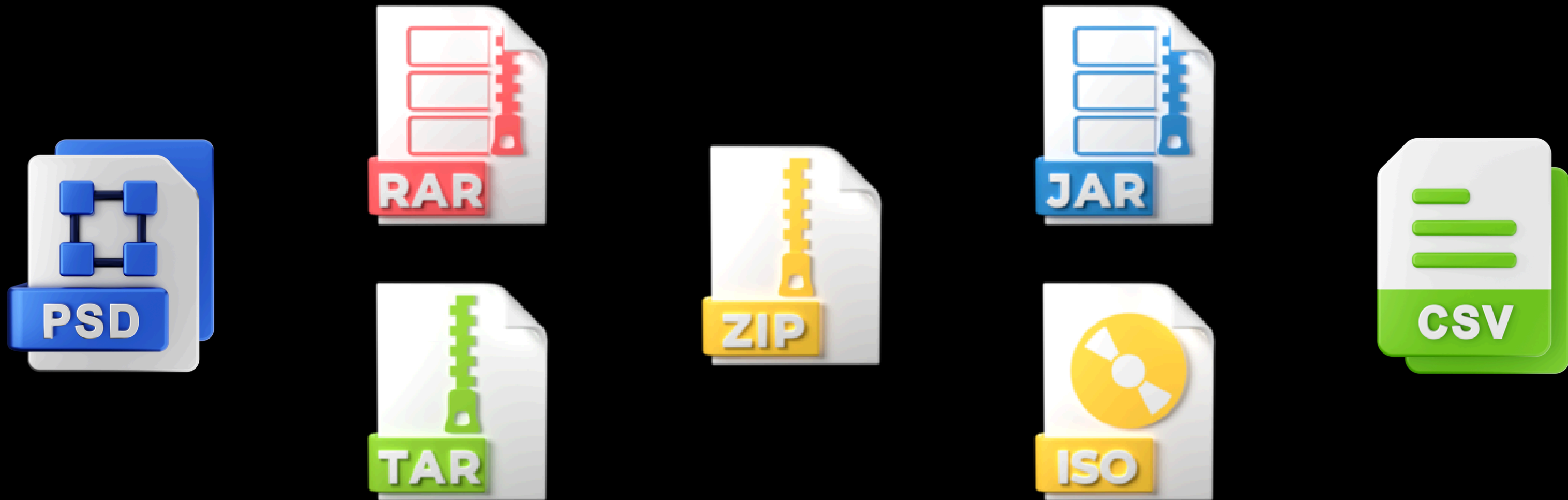
- Logs
- Memory dumps
- Packet dumps
- Files

Artifact: any traces left behind that reveal user or process activity

File Forensics

File Formats

Files are structured with a specific pattern defined by
its format



Specifications are often posted online as RFCs

file command

You can use the linux 'file' command to view the file format (it checks the magic numbers)

```
└─(19:43:41)─> xxd mypedf.pdf | head
00000000: 2550 4446 2d31 2e36 0d25 e2e3 cfd3 0d0a %PDF-1.6.%.....
00000010: 3230 3731 2030 206f 626a 0d3c 3c2f 4c69 2071 0 obj.<</Li
00000020: 6e65 6172 697a 6564 2031 2f4c 2038 3431 nearized 1/L 841
00000030: 3637 362f 4f20 3230 3733 2f45 2031 3338 676/O 2073/E 138
00000040: 3636 372f 4e20 332f 5420 3834 3039 3330 667/N 3/T 840930
00000050: 2f48 205b 2039 3630 2031 3538 325d 3e3e /H [ 960 1582]>>
00000060: 0d65 6e64 6f62 6a0d 2020 2020 2020 2020 .endobj.
00000070: 2020 0d0a 3231 3531 2030 206f 626a 0d3c ..2151 0 obj.<
00000080: 3c2f 4465 636f 6465 5061 726d 733c 3c2f </DecodeParms<</
00000090: 436f 6c75 6d6e 7320 352f 5072 6564 6963 Columns 5/Predic
```

```
└─~/Downloads
└─(19:43:43)─> file mypedf.pdf
mypedf.pdf: PDF document, version 1.6 (zip deflate encoded)
```

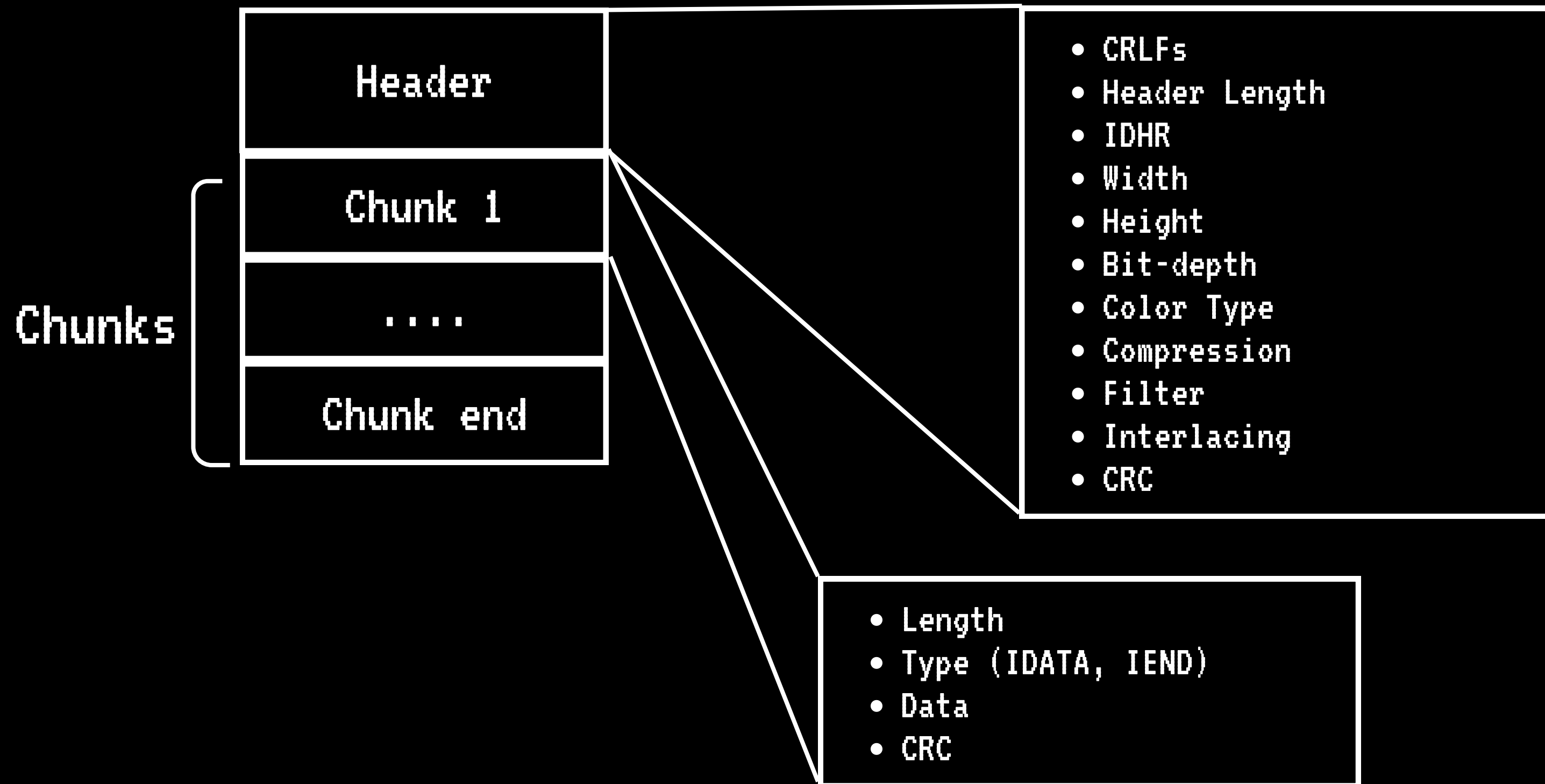
```
└─(19:43:49)─> xxd mypedf2.pdf | head
00000000: 4344 462d 312e 360d 25e2 e3cf d30d 0a32 CDF-1.6.%.....2
00000010: 3037 3120 3020 6f62 6a0d 3c3c 2f4c 696e 071 0 obj.<</Lin
00000020: 6561 7269 7a65 6420 312f 4c20 3834 3136 earized 1/L 8416
00000030: 3736 2f4f 2032 3037 332f 4520 3133 3836 76/O 2073/E 1386
00000040: 3637 2f4e 2033 2f54 2038 3430 3933 302f 67/N 3/T 840930/
00000050: 4820 5b20 3936 3020 3135 3832 5d3e 3e0d H [ 960 1582]>>.
00000060: 656e 646f 626a 0d20 2020 2020 2020 2020 endobj.
00000070: 200d 0a32 3135 3120 3020 6f62 6a0d 3c3c ..2151 0 obj.<<
00000080: 2f44 6563 6f64 6550 6172 6d73 3c3c 2f43 /DecodeParms<</C
00000090: 6f6c 756d 6e73 2035 2f50 7265 6469 6374 olumns 5/Predict
```

```
└─~/Downloads
└─(19:43:56)─> file mypedf2.pdf
mypedf2.pdf: data
```

If file format is not expected, then file command defaults to raw data

File Formats - PNG

PNG



PNG images are also often compressed

File Formats - PNG

PNG

- CRLFs
- Length
- IDHR
- Width
- Height
- Bit-depth
- Color Type
- Compression
- Filter
- Interlacing
- CRC

Chunk 1:

- Length
- Type
- Data
- CRC

....

Chunk end

Header

PNG Header CRLFS

89 50 4e 47 0d 0a 1a 0a

Length

00 00 00 0d

IHDR

49 48 44 52

Bitdepth, Ctype, Compression,
Filter, Interlacing

Width

00 00 04 28

Height

00 00 02 44

CRC

d4 8e 15 e7

Chunk 1

Chunk length Chunk Type (IDAT)

00 03 41 cd

49 44 41 54

Chunk Data

78 9C EC BD ... DC EC 73 18

CRC

36 43 a3 9a

Chunk End

Chunk length Chunk Type (IEND)

00 00 00 00

49 45 4E 44

CRC

AE 42 60 82

File Formats - PNG

How can we abuse the PNG file format to hide data?

- Stuff data past the IEND chunk
- Image Steganography (we'll talk about this later)

You can use `pngcheck` to see if a PNG is normal

```
$ pngcheck image.png  
OK: image.png (1920x1080, 32-bit RGB+alpha, non-interlaced, 96.6%).
```

Sidenote - Cyclic Redundancy Check

A short error detecting hash of 8bit, 16bit, 32bit size

Checks integrity of data and if it was modified during transit



`reveng` can be used to calculate CRC parameters from samples

Sidenote - Magic Numbers

Collections of bytes with no real meaning

THEY ARE MAGIC

* Comprehensive list: [https://
en.wikipedia.org/wiki/
List_of_file_signatures](https://en.wikipedia.org/wiki/List_of_file_signatures)

Often used by files as a
shorthand for file detection



- PNG files: 89 50 4e 47 → .PNG
- PDF files: 25 50 44 46 → %PDF
- PE files: 4d 5a → MZ

ZIP Files

Contains:

Records:

- File data compressed with DEFLATE

Dir Entries:

- Describes more data about records

Defined by the End of Central
Directory section

ZIP

Record 1

...

Record n

Dir Entry 1

...

Dir Entry n

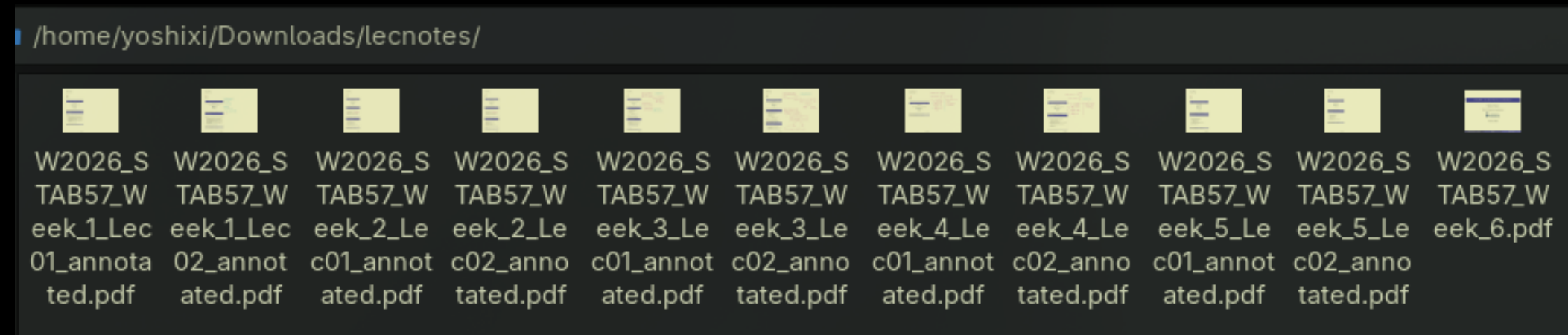
End of Central
Directory

Records

Central
Directory

* at the end of file. not a header, made to be easier to expand a zip

ZIP Files

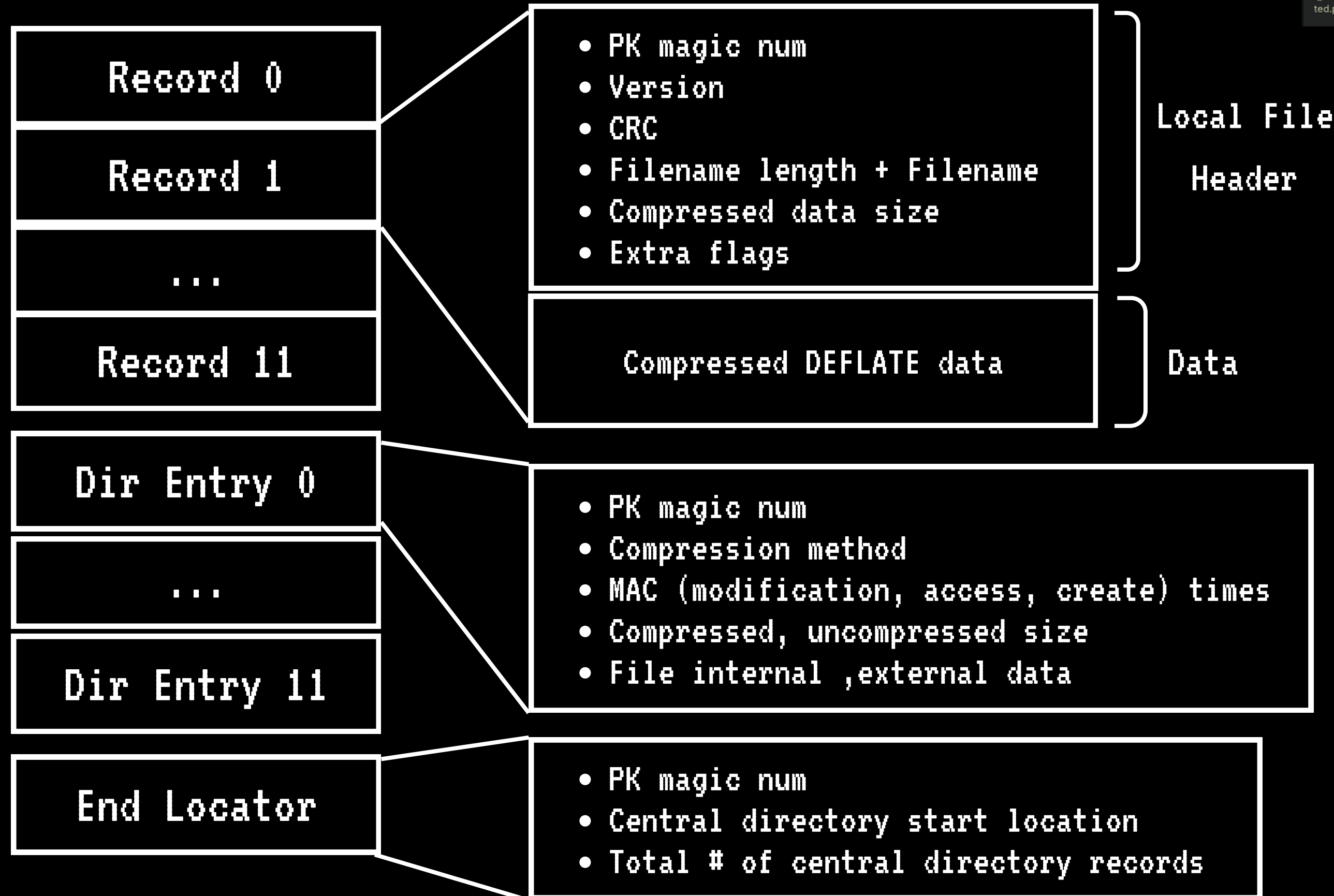
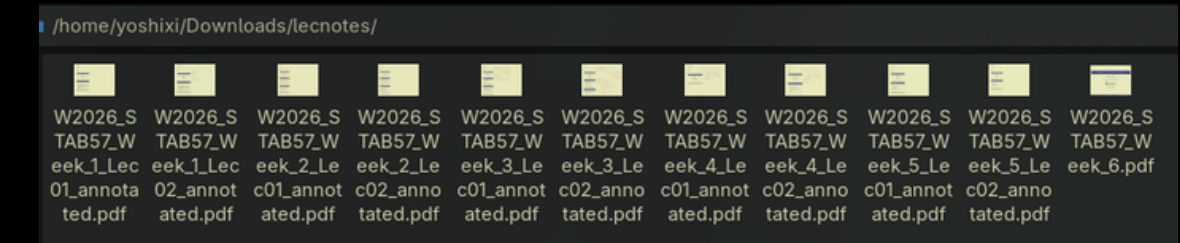


```
(09:55:04)→ zip -r lecnotes.zip lecnotes
adding: lecnotes/ (stored 0%)
adding: lecnotes/W2026_STAB57_Week_5_Lec01_annotated.pdf (deflated 12%)
adding: lecnotes/W2026_STAB57_Week_4_Lec01_annotated.pdf (deflated 8%)
adding: lecnotes/W2026_STAB57_Week_2_Lec01_annotated.pdf (deflated 10%)
adding: lecnotes/W2026_STAB57_Week_2_Lec02_annotated.pdf (deflated 9%)
adding: lecnotes/W2026_STAB57_Week_6.pdf (deflated 4%)
adding: lecnotes/W2026_STAB57_Week_1_Lec02_annotated.pdf (deflated 8%)
adding: lecnotes/W2026_STAB57_Week_4_Lec02_annotated.pdf (deflated 9%)
adding: lecnotes/W2026_STAB57_Week_1_Lec01_annotated.pdf (deflated 8%)
adding: lecnotes/W2026_STAB57_Week_3_Lec01_annotated.pdf (deflated 9%)
adding: lecnotes/W2026_STAB57_Week_3_Lec02_annotated.pdf (deflated 9%)
adding: lecnotes/W2026_STAB57_Week_5_Lec02_annotated.pdf (deflated 13%)
```

ZIP Files

* All other records are kind of similar, but some are directories with no data like record 0 (/lectures)

← directory



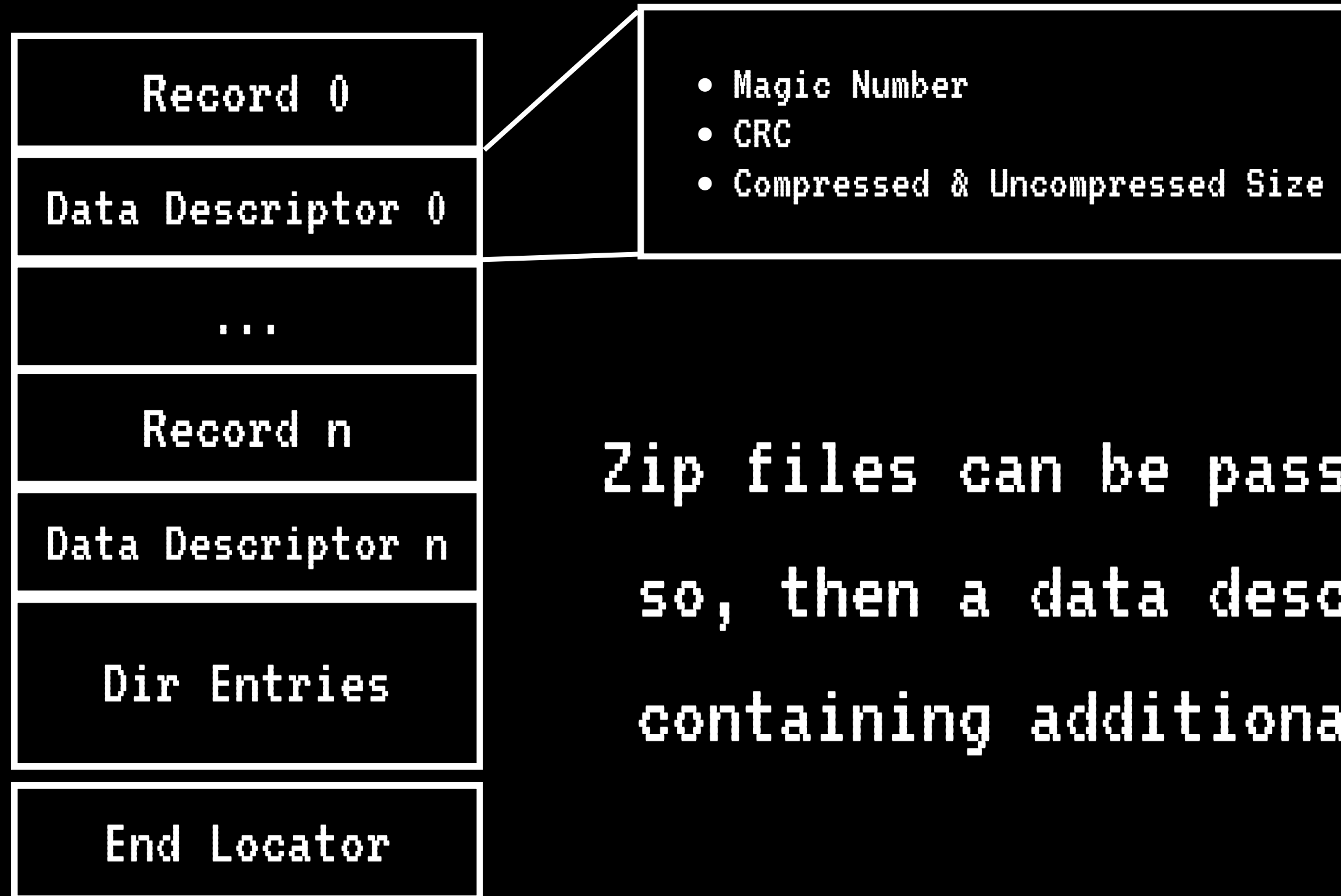
ZIP Parsers

1. Read the End of Central Directory section to find the first dir entry
2. Read the first dir entry, then go to record
3. Go to next dir entry until we read all dir entries

The format is 'abused' sometimes:

- Self extracting zips (WinZip) make the first bytes of the file a PE executable
- You can stuff any data between each chunk
- Change the end of central directory to hide files

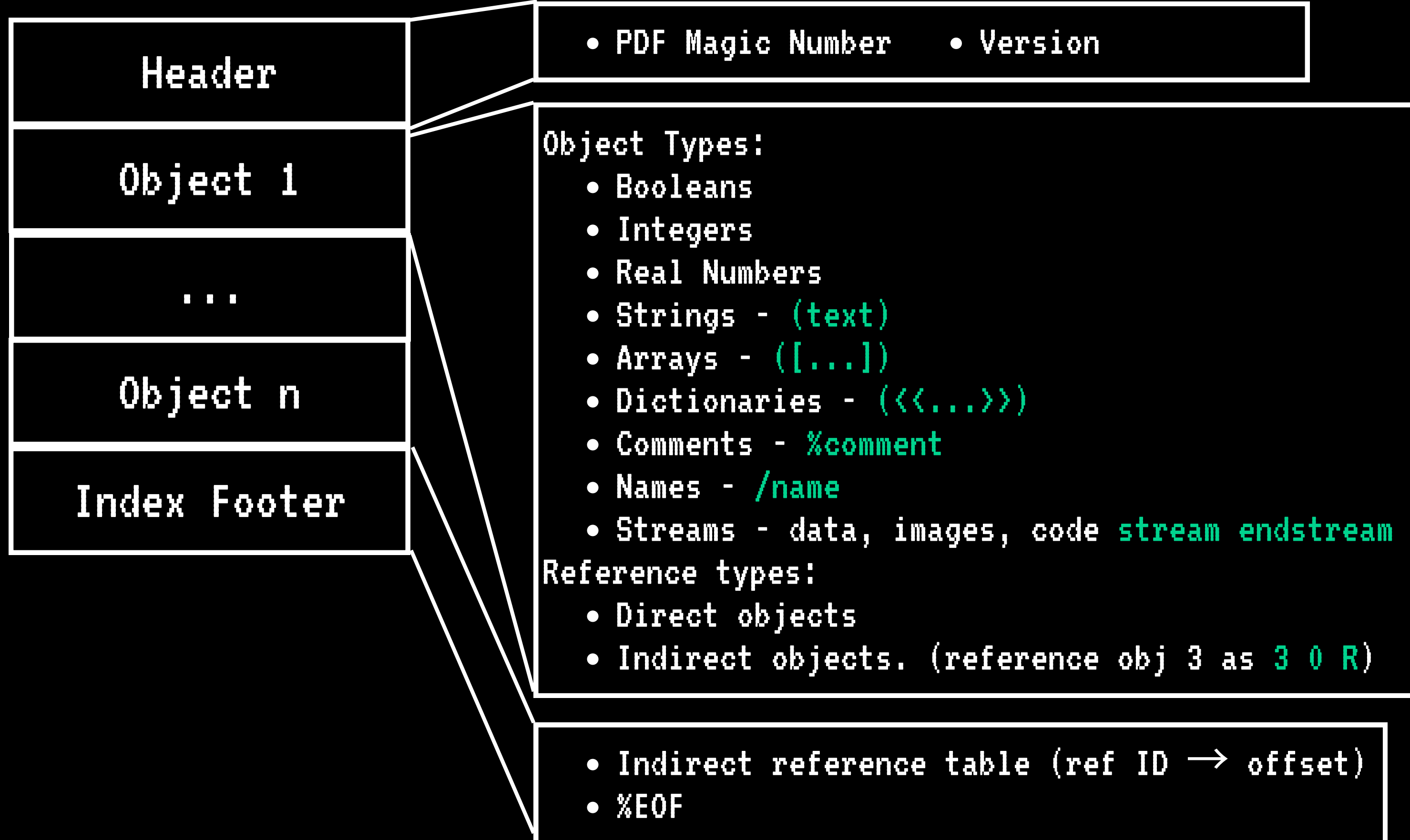
ZIP Files



Zip files can be password protected too. If so, then a data descriptor block is added containing additional data for decryption

PDF Files

COS
Tree
Body



PDF Files

PDF is one of those weird filetypes where
you can read portions of it like plaintext

```
1 %PDF-1.6^M%âãÿÓ^M
2 2071 0 obj^M<</Linearized 1/L 841676/O 2073/E 138667/N 3/T 840930/H [ 960 1582]>>^Mendobj^M
3 2151 0 obj^M<</DecodeParms<</Columns 5/Predictor 12>>/Filter/FlateDecode/ID[<EB7E51C5A927894FB83C94A7A18A21BB>
  /Info 2070 0 R/Length 197/Prev 840931/Root 2072 0 R/Size 2267/Type/XRef/W[1 3 1]>>stream^M
4 hþbbd`^P`^b`à¼^G"^Y÷<83>H!;`öMì$ÃM|²Ä«; <85>JçLû^0&cÁ$^K<98><8c>^G<93>`_3AØL`2^AD:÷<83>H¾É`ÛM`ò^U<88>d^C³^Yo<80
  <99><87>p^C<83>%<98><9d>^0$^Y£ó@l9^U^P©<9c>^E&x<80>HAS^P)©^FV<9f>^H"eçÃ¸v<80>É^X^P)ê^0g3ê?^F<92>ÿ<9e>.d`bdà,^
5 endstream^Mendobj^Mstartxref^M
6 0^M
7 %%EOF^M
8
9 2266 0 obj^M<</C 2079/Filter/FlateDecode/I 2101/Length 1475/O 1611/S 758/V 1627>>stream^M
```

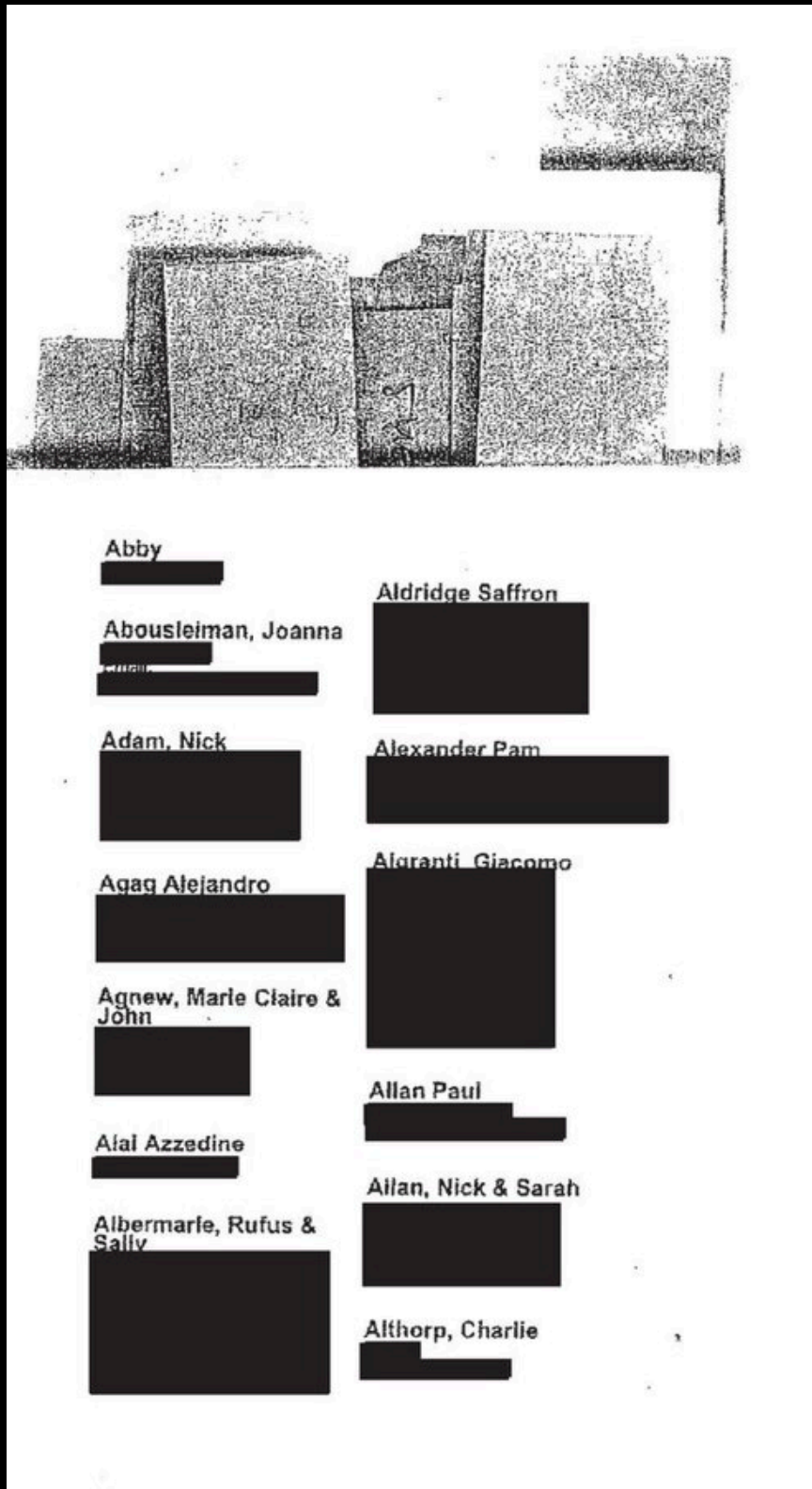
still use a hex editor though

Epstein Files PDF Artifact Removal

When epstein file documents were initially released, black boxes were added to cover up text.

These are just PDF objects that could be removed or bypassed via full document copy paste

You can also use `pdftotext`



PDF File Decompression

```
qpdf --stream-data=uncompress input.pdf output.pdf
```

```
~/Downloads/prettydevilishfile david arch:pts/0  
(11:52:30) → qpdf pretty_devilish_file.pdf output.pdf 2 ↵ (Sat, Oct 18)  
WARNING: pretty_devilish_file.pdf: file is damaged  
WARNING: pretty_devilish_file.pdf: can't find startxref  
WARNING: pretty_devilish_file.pdf: Attempting to reconstruct cross-reference table  
WARNING: pretty_devilish_file.pdf (trailer, offset 1412): dictionary has duplicated key /Root; last occurrence overrides earlier ones  
WARNING: pretty_devilish_file.pdf (object 7 0, offset 1402): expected endobj  
WARNING: pretty_devilish_file.pdf (trailer, offset 1410): invalid /ID in trailer dictionary  
WARNING: pretty_devilish_file.pdf, trailer at offset 1412 -> dictionary key /ID: operation for array attempted on object of type null: returning null  
WARNING: pretty_devilish_file.pdf, trailer at offset 1412 -> dictionary key /ID -> null returned from invalid array access: operation for string attempted on object of type null: returning empty string  
WARNING: pretty_devilish_file.pdf (object 4 0, offset 915): expected endobj  
qpdf: operation succeeded with warnings; resulting file may have some problems  
~/Downloads/prettydevilishfile david arch:pts/0
```

Executable File Format

There are also a variety of file formats for executable programs:

PE:



- For Windows
- Used in
 - .exe
 - .dll

ELF:



- For Linux
- Used in
 - .bin
 - .so

Go check out the rev-eng seminar for more info

Probing Files with Hex Editors

```
0:0000  25 50 44 46 2D 31 2E 36 0D 25 E2 E3 CF D3 0D 0A %PDF-1.6.%âãĬŎ..
0:0010  32 30 37 31 20 30 20 6F 62 6A 0D 3C 3C 2F 4C 69 2071|0|obj|. <</Li
0:0020  6E 65 61 72 69 7A 65 64 20 31 2F 4C 20 38 34 31 nearized 1/L 841
0:0030  36 37 36 2F 4F 20 32 30 37 33 2F 45 20 31 33 38 676/O 2073/E 138
0:0040  36 36 37 2F 4E 20 33 2F 54 20 38 34 30 39 33 30 667/N 3/T 840930
0:0050  2F 48 20 5B 20 39 36 30 20 31 35 38 32 5D 3E 3E /H [ 960 1582]>>
0:0060  0D 65 6E 64 6F 62 6A 0D 20 20 20 20 20 20 20 20 .endobj.
0:0070  20 20 0D 0A 32 31 35 31 20 30 20 6F 62 6A 0D 3C ..2151 0 obj.<
0:0080  3C 2F 44 65 63 6F 64 65 50 61 72 6D 73 3C 3C 2F </DecodeParms<</
0:0090  43 6F 6C 75 6D 6E 73 20 35 2F 50 72 65 64 69 63 Columns 5/Predic
0:00A0  74 6F 72 20 31 32 3E 3E 2F 46 69 6C 74 65 72 2F tor 12>>/Filter/
0:00B0  46 6C 61 74 65 44 65 63 6F 64 65 2F 49 44 5B 3C FlateDecode/ID[<
0:00C0  45 42 37 45 35 31 43 35 41 39 32 37 38 39 34 46 EB7E51C5A927894F
0:00D0  42 38 33 43 39 34 41 37 41 31 38 41 32 31 42 42 B83C94A7A18A21BB
```

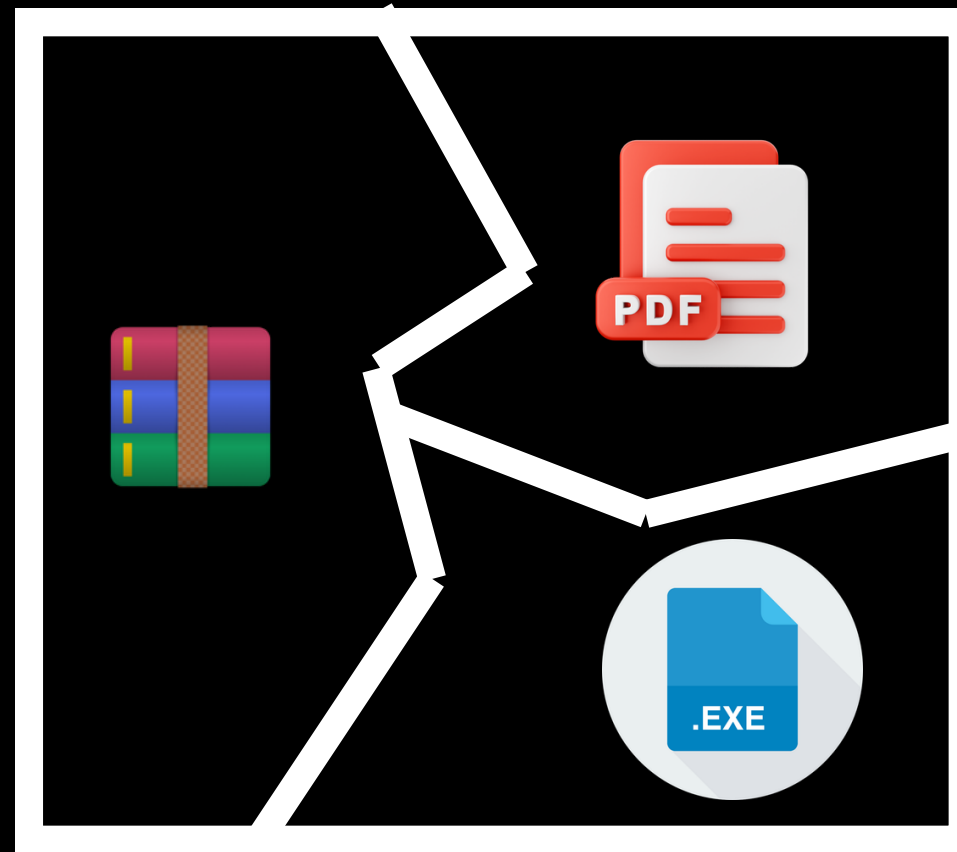
Template Results - PDF.bt [↻](#)

Name	Value	Start	Size	Type	Color	Comment
▶ PDF File Header comment	%PDF-1.6	0h	8h	struct PDFHe...		Valid PDF version
▶ PDF Body Section		8h	CF964h	struct PDFBo...		
▶ Conventional cross reference ...		CF96Ch	0h	struct PDFxre...		
▶ startxref with offset[0]	startxref 0	246h	Ch	struct PDFSta...		startxref offset 0 was not 'xref': w..
▶ %%EOF end of file comment[0] %%EOF		254h	5h	struct PDFOF		
▶ Conventional cross reference ...		259h	0h	struct PDFxre...		
▶ startxref with offset[1]	startxref 116	CD7B5h	Eh	struct PDFSta...		startxref offset 116 was not 'xref':.
▶ %%EOF end of file comment[1] %%EOF		CD7C5h	5h	struct PDFOF		
▶ Conventional cross reference ...		CD7CAh	0h	struct PDFxre...		
▶ startxref with offset[2]	startxref 849937	CF96Dh	10h	struct PDFSta...		startxref offset 849937 was not 'x..
▶ %%EOF end of file comment[2] %%EOF		CF97Eh	5h	struct PDFOF		

- ImHex
- HxD
- 010 Editor

Polyglots

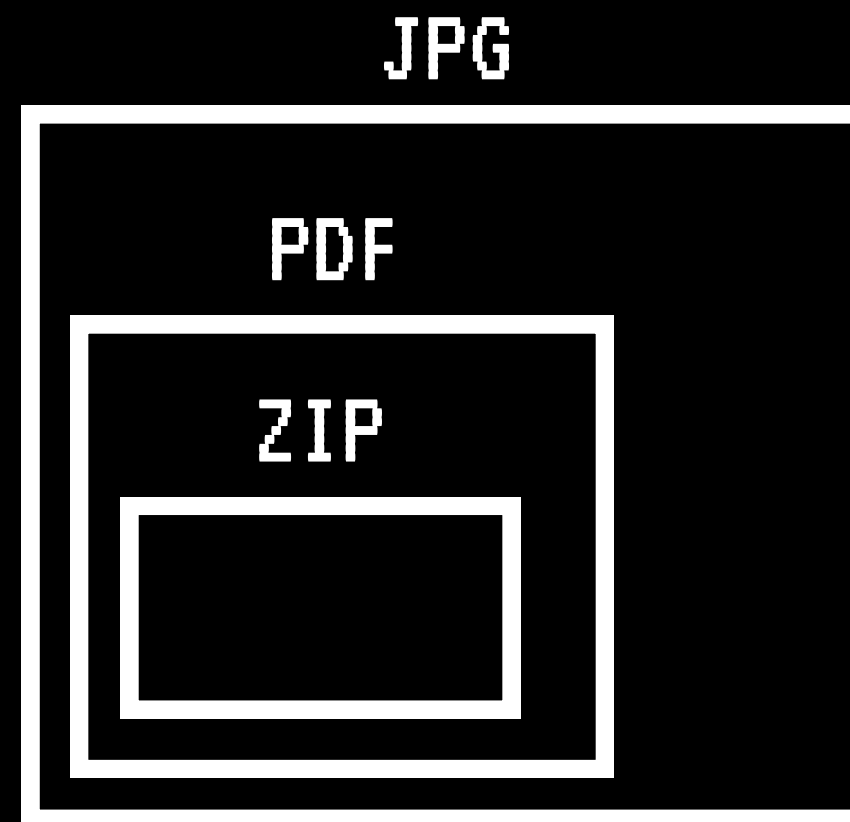
A file with multiple formats



You can identify a polyglot with `file -k`

Embedded Files

Sometimes files can be embedded into other files



Like a polyglot, but the inside file is not meant to be parsed

```
steghide embed --coverfile <coverinput> --embedfile <embedinput>
```

File Carving

The process of carving out these embedded files

```
foremost -i <input_file> -o recovered/
```

```
binwalk -e <input_file>
```

```
~/Downloads/tempdir _____  
(14:52:31)→ ls  
📄 flag2of2-final.pdf  
~/Downloads/tempdir _____  
(14:52:33)→ foremost -i flag2of2-final.pdf -o recovered  
Processing: flag2of2-final.pdf  
|*|  
~/Downloads/tempdir _____  
(14:52:34)→ ls recovered/  
📄 audit.txt 📄 png  
~/Downloads/tempdir _____  
(14:52:37)→ ls recovered/png  
📄 00000000.png
```

File Carving

You can also manually carve out a file

```
dd if=<input_file> of=<output_file> bs=1 skip=<num_blocks> count=<num_count>
```

- bs: block size
- skip: the number of blocks to skip
- count: the number of blocks to copy

Polyglot Demo!

Lets try to recover information in this mangled file

<https://ctfd.yoshixi.net/challenges#miforensiks-8>

Metadata

Data about data. Includes things like:

- Location
- Access times
- Filesize
- ...

Can be:

- Embedded within a file
- Stored in the file system

Embedded Metadata

This is metadata embedded within the file
(often in the file headers)

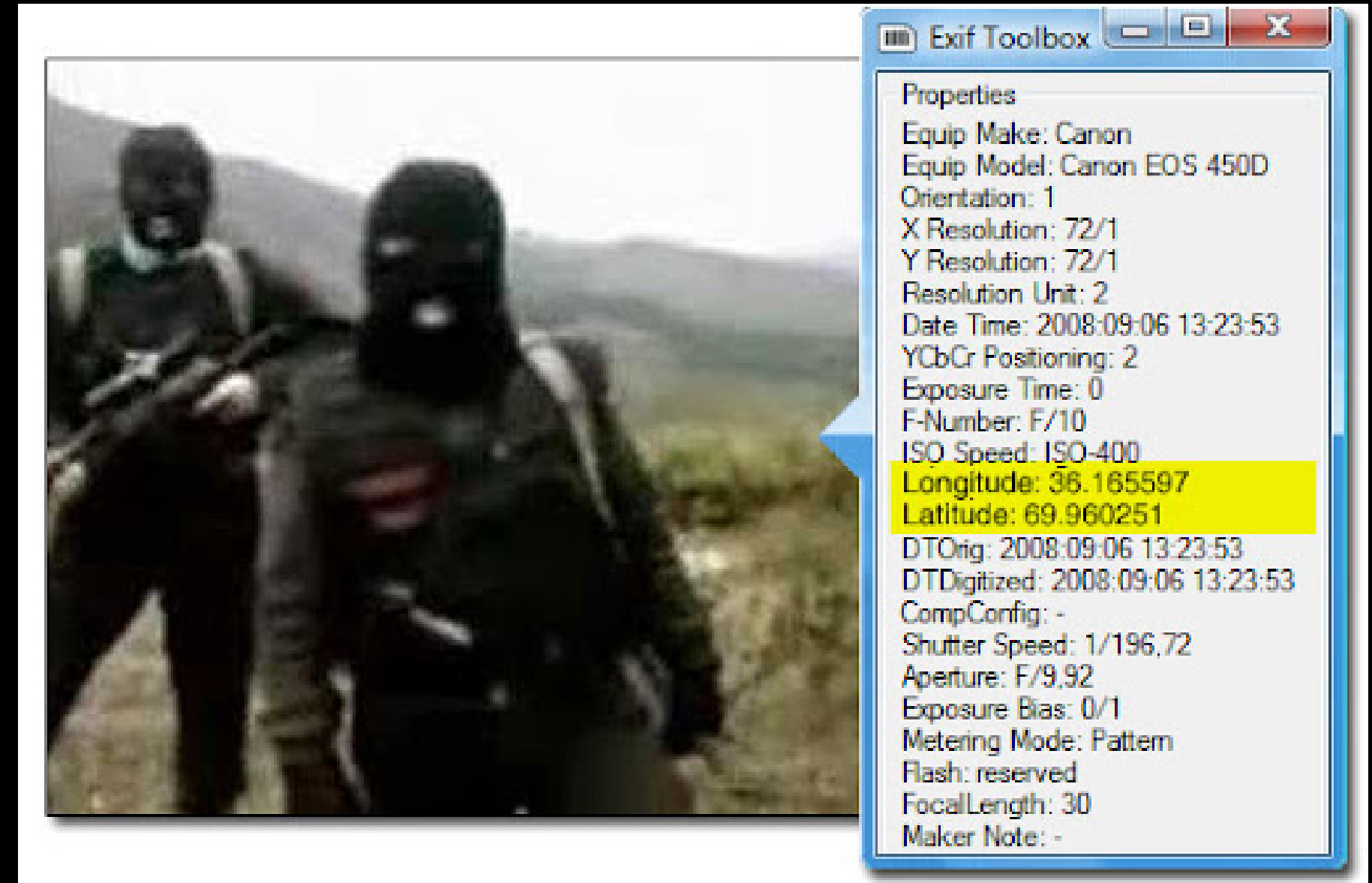
- EXIF: Images, Videos, Audio
- XMP: Images
- IPTC: Images
- MXF: Images
- ID3: Audio
- ... so much more formats

EXIF Metadata

Common metadata in compatible
with a variety images, video,
audio file formats

Contains:

- Camera settings
- Image size metrics
- Date and time
- Location (GPS) information
- ...



* social media sites often
strip some/all metadata

Exiftool

Very useful tool that decodes EXIF metadata,
IPTC, variety of other metadata formats

```
yoshixi@DESKTOP-SG1D0R0:~$ exiftool WIN_20260324_17_10_54_Pro.jpg
ExifTool Version Number      : 12.76
File Name                    : WIN_20260324_17_10_54_Pro.jpg
Directory                    : .
File Size                    : 95 kB
File Modification Date/Time  : 2026:03:24 17:10:55-04:00
File Access Date/Time       : 2026:03:24 17:12:31-04:00
File Inode Change Date/Time  : 2026:03:24 17:11:38-04:00
File Permissions             : -rw-r--r--
File Type                    : JPEG
File Type Extension         : jpg
MIME Type                    : image/jpeg
JFIF Version                 : 1.01
Resolution Unit              : inches
X Resolution                  : 96
Y Resolution                  : 96
Exif Byte Order              : Big-endian (Motorola, MM)
Software                     : Windows 11
Date/Time Original           : 2026:03:24 17:10:54
Sub Sec Time Original        : 566
Padding                      : (Binary data 4108 bytes, use -b option to extract)
Image Width                  : 1280
Image Height                 : 720
Encoding Process             : Baseline DCT, Huffman coding
Bits Per Sample              : 8
Color Components              : 3
```

Filesystem Metadata

This is metadata stored by the filesystem
about files/directories.

- Inode tables (Linux ext4)
- MFT B+ tree (Windows NTFS)
- B tree catalog (macOS APFS)
- ...

ext4 Filesystem Inodes

Inode

- Owner UID, GID
- Size
- MAC times
- Permissions

Direct blocks

1 indirect

2 indirect

3 indirect

Metadata

Block ptrs

ext4 filesystem uses Inodes to store information about files

- Owners
- Permissions
- MAC (modified, access, change) times
- Creation time

You can read this metadata with the

`stat` command/syscall

Sidenode - Timestamps

Timestamps are often records as unix timestamps

Unix timestamp: seconds from Jan 01 1970

Modified, Accessed, Created (MAC) timestamps:

- Modified (mtime): file data changed
- Access (atime): access/read file
 - Often always changes with mtime
 - Often set to off (reading files is too common, so expensive)
- Change (ctime): filesystem metadata changed



ext4 Filesystem xattrs

Additional custom metadata for a file.

name : value

Used often for ACL, file hashes

can be read with `stat` or `getfattr`

* xattrs are very common in macOS too

Steganography

Steganography

The process of embedding secret messages
within the data of files.



Viewable Message



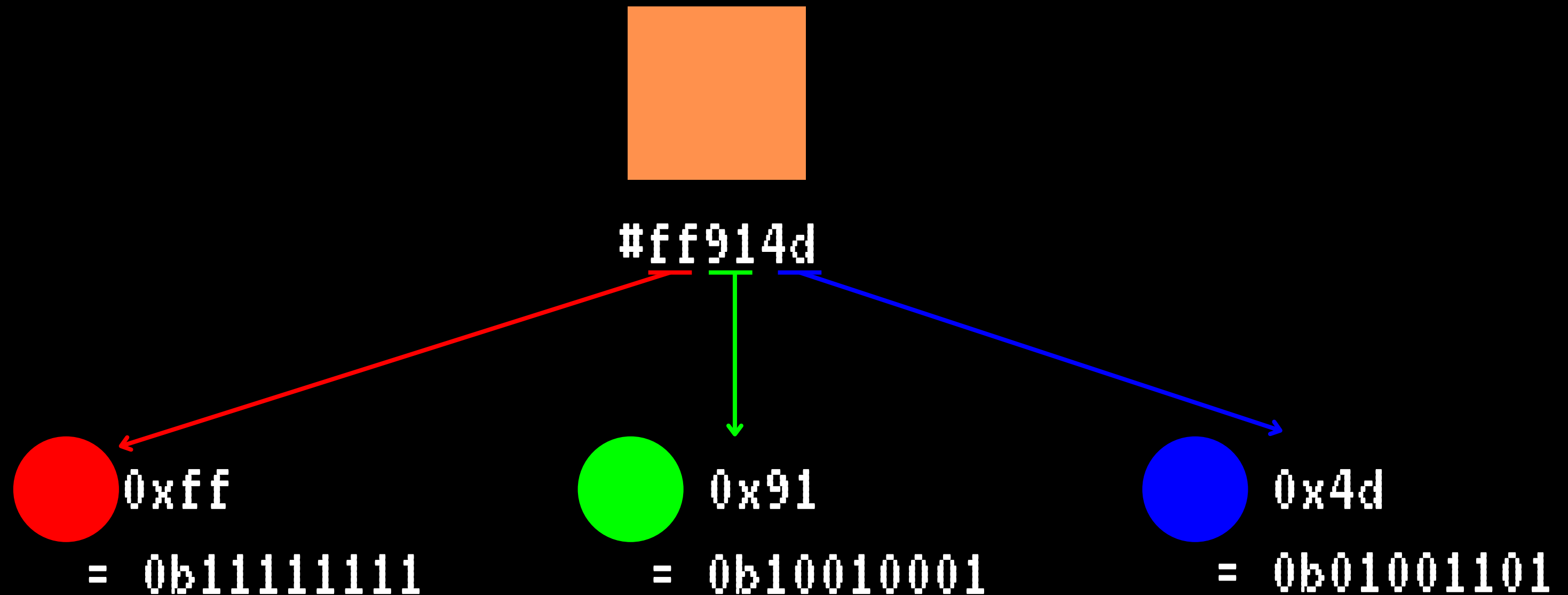
**“Meet me at the
park tonight at
10pm”**

Hidden Message

Bit Planes

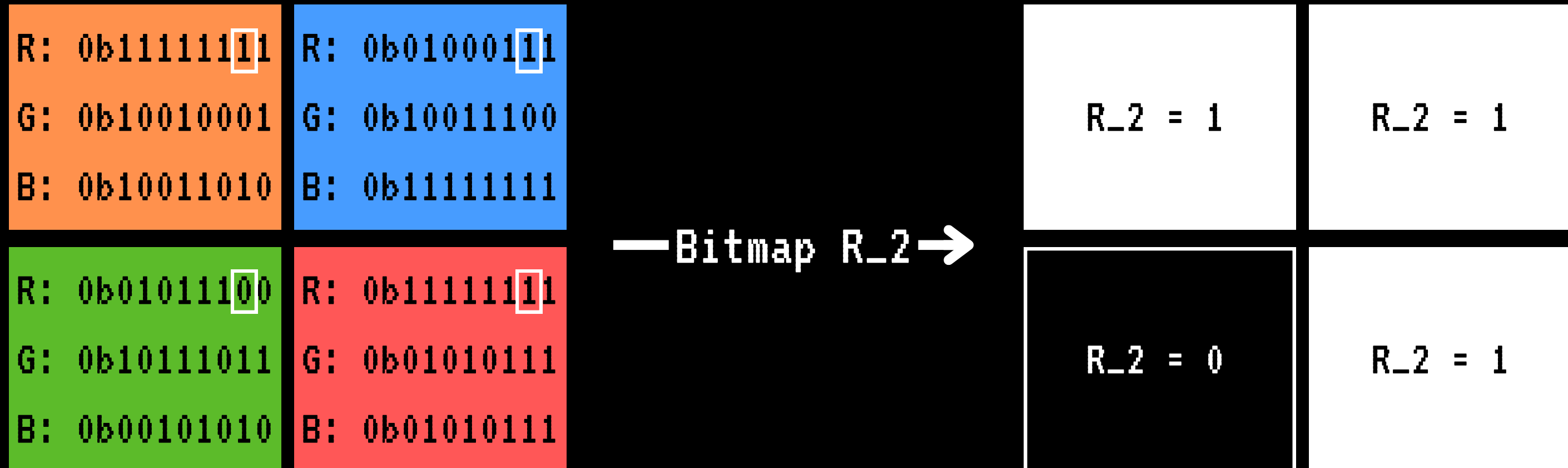
A common way to represent images is RGB-255 color mode

Each pixel is comprised of a combination of RGB values



Bit Planes

We represent an image as a n by m dimensional matrix of these
RGB-255 values

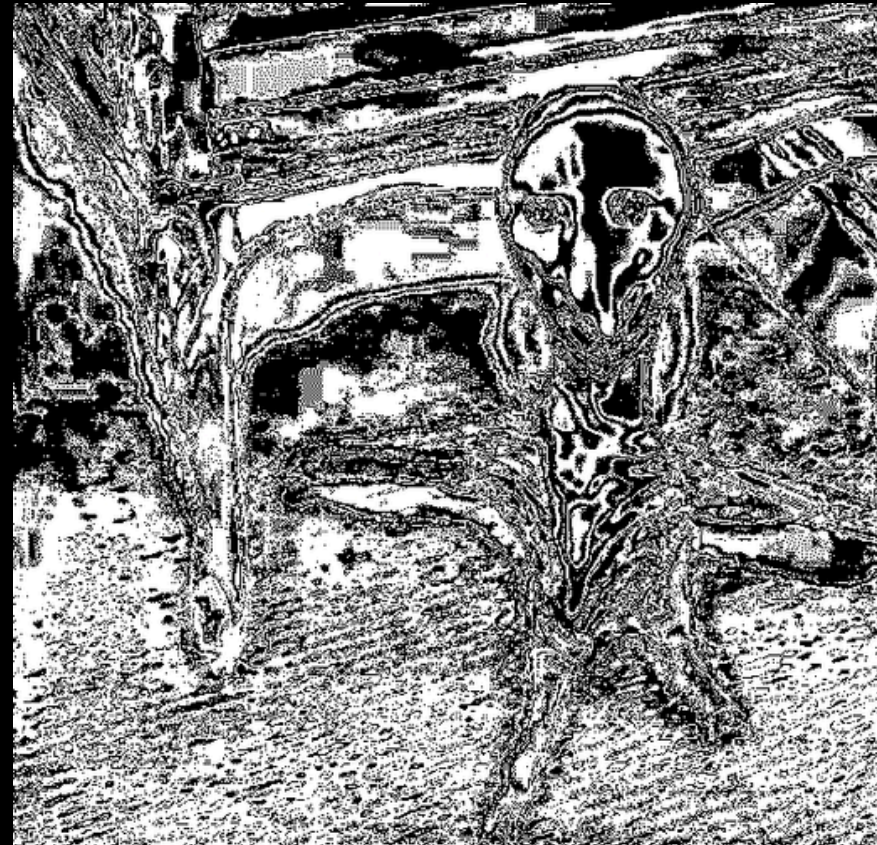


A bitplane image would be a n by m matrix comprised of only a
single bit value from either the R, G, or B channel

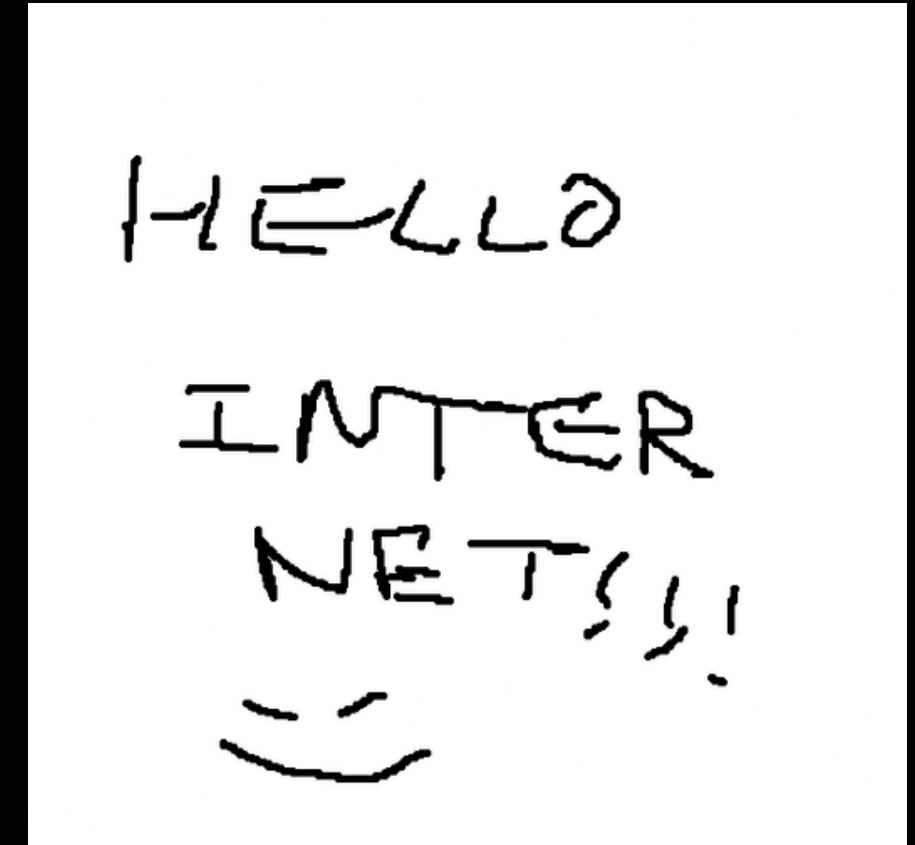
Bit Plane / LSB Steganography



Original Image



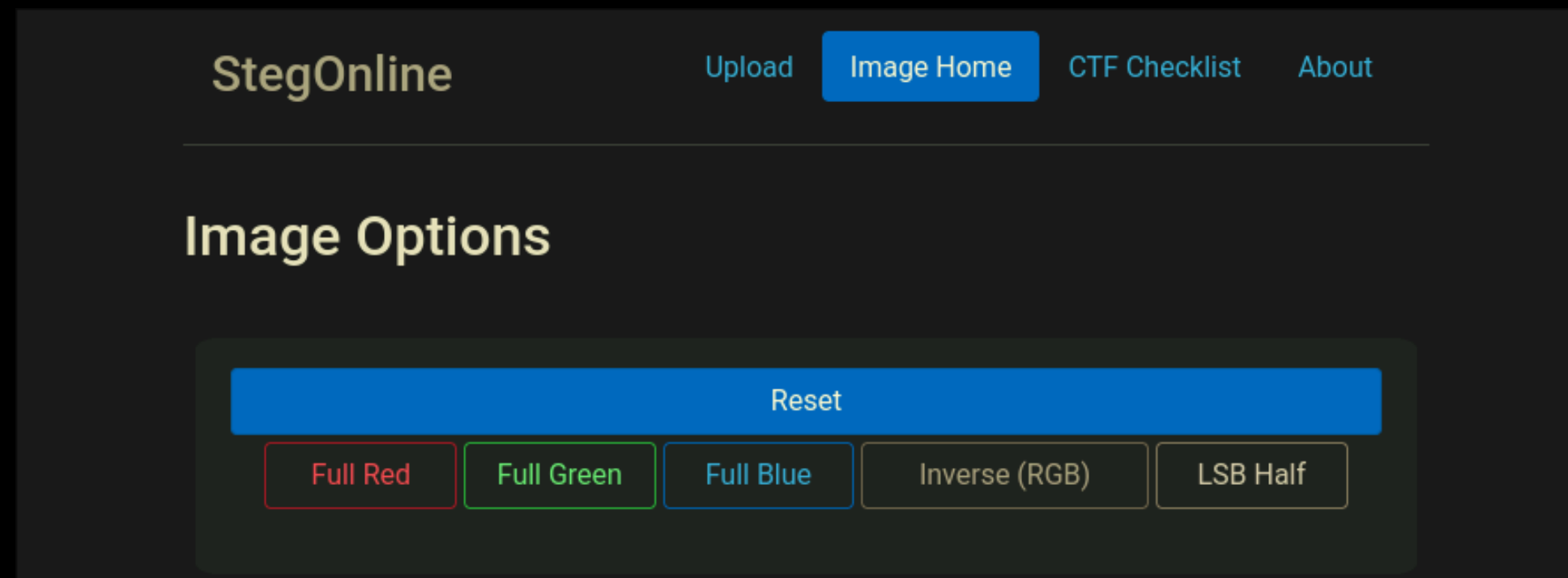
R1 bitmap



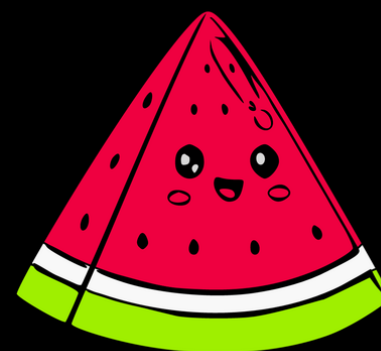
G2 bitmap
(i modified this)

StegOnline & Aperisolve

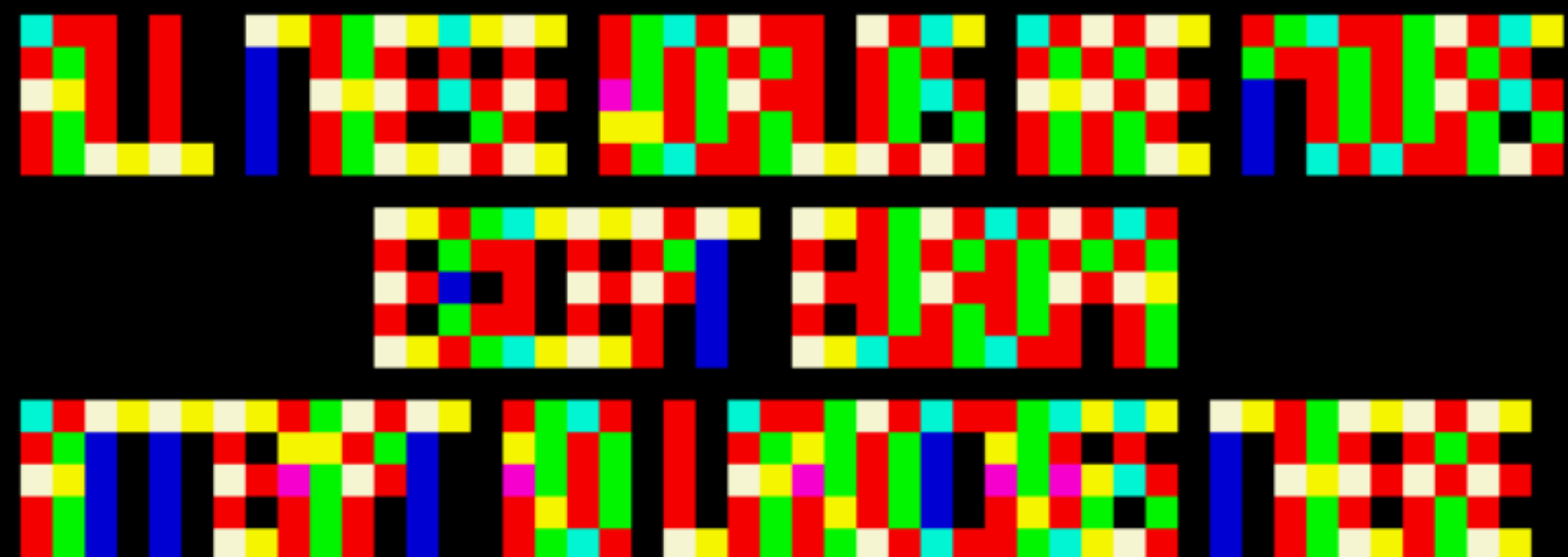
You can use the tool StegOnline to browse bit planes.



Aperisolve also has a variety of LSB tools



Subpixel Steganography

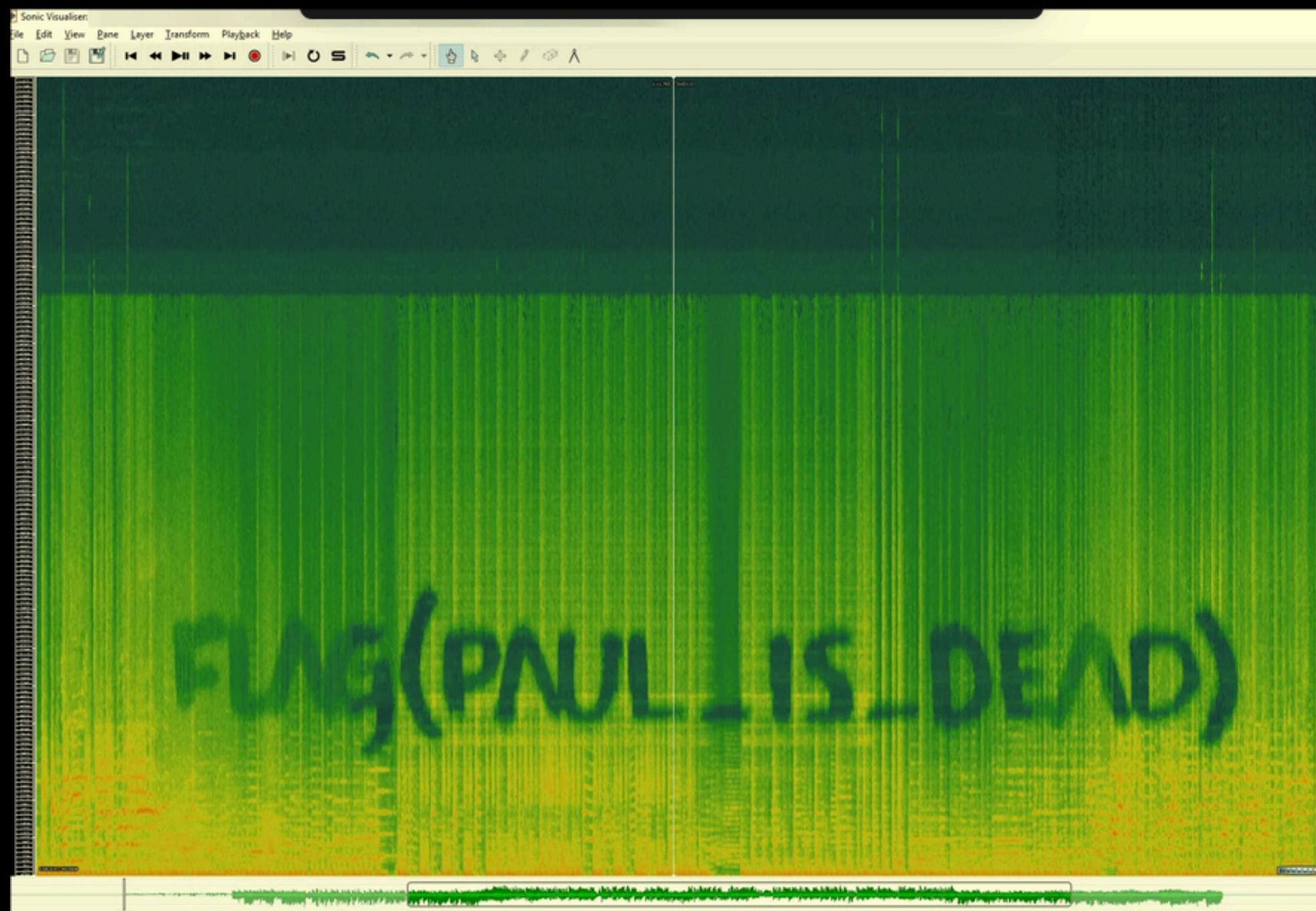


And the message revealed:



Spectrograms

Viewing the waveform diagrams of audio files to reveal shapes



You can use Sonic Visualizer

Slow Scan Television

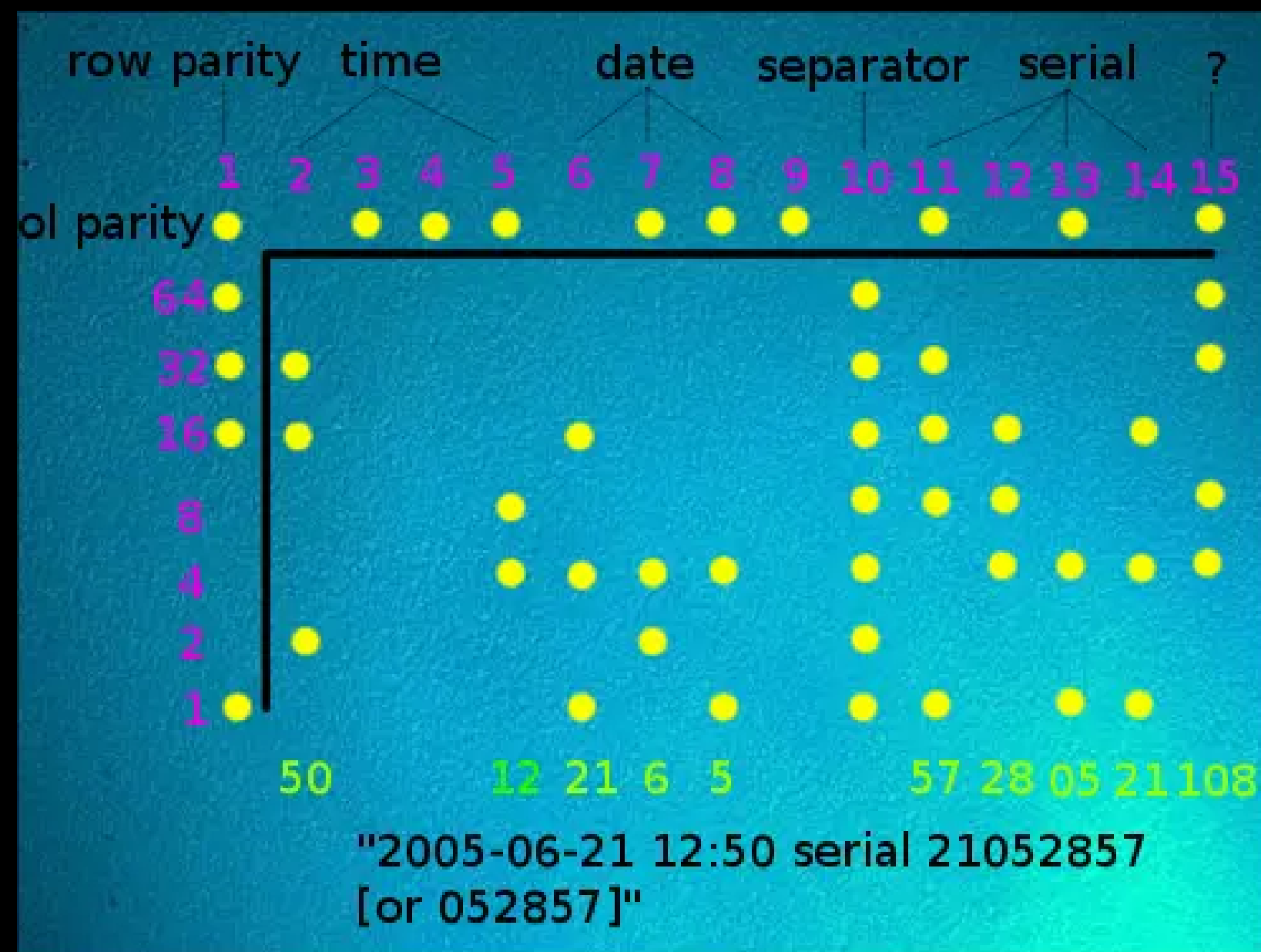
An encoding amateur radio operators use to transmit and receive images over radio.



You can decode with QSSTV

Printer Dots

Laser printers print a encoded machine identification code for each paper printed. Naked to invisible eye, but visible under blue light.



Can be used to catch counterfeiters :)

Network Forensics

Computer Network

The hardware, software and protocols that allow communication between two different computers



Host 1: 192.168.97.22

Host 2: 192.168.97.45

The way packets are transferred differs based off scope:

- Local (LAN)
- Internet (WAN)

Packets & Protocols

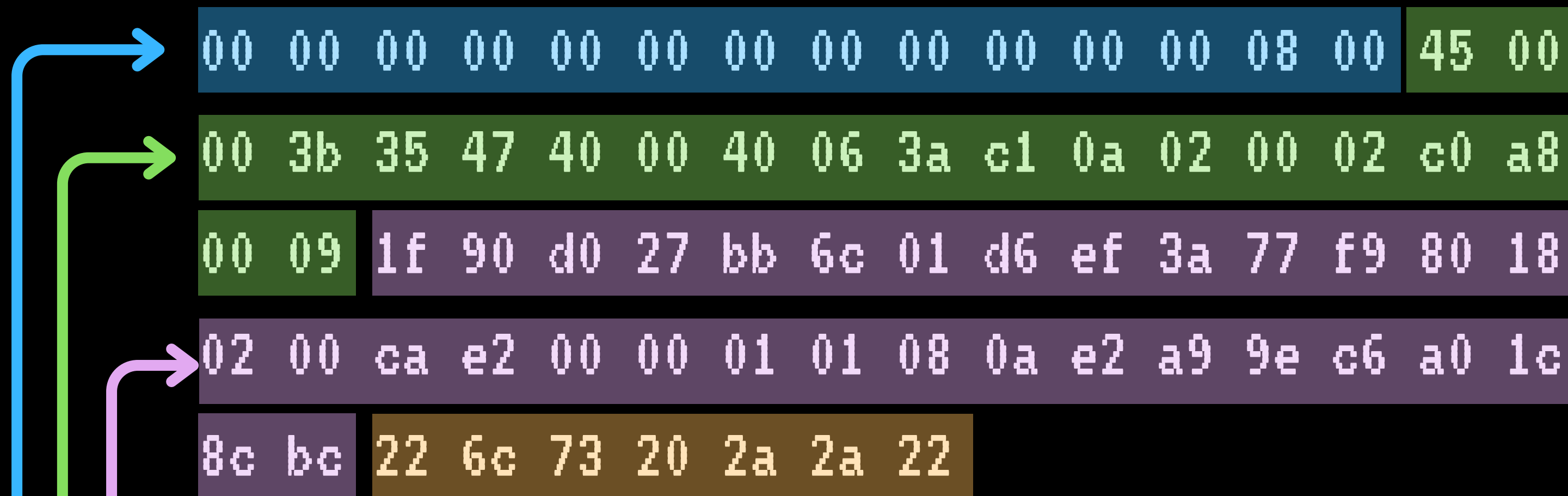
Packets/Frames are structured data sent across a network.

They contain:

- Raw payload data
- Metadata for each protocol required for transit



Example Packet Data



Ethernet Header

IP Header

TCP Header

Payload

Decodes to "ls **"

OSI Model

Network protocols are created for different levels of the OSI models

Application	: HTTP, FTP, BitTorrent, RDP, SSH, ...
Presentation	: TLS/SSL, ...
Session	: SOCKS, RPC, SMB, ...
Transport	: TCP, UDP, QUIC, ...
Network	: IP, ARP, ICMP, ...
Data Link	: IEE 802.11, Ethernet, ...
Physical	: CAN bus, Ethernet physical, ...

Network Interface

In order to be connected to a network, you must have an interface to the network for the OS to work with.

These network interfaces could be:

- Physical network interface (NIC)
- Virtual network interface (VMs, VPNs for VLAN, Loopback)

* The # of network interfaces you have affects the # of local networks you can join

If an interface needs it, it is given its own MAC address

Sidenote - MAC Address

Addresses used for network interfaces that will be used for data-link layer (OSI layer 2) communication

54:27:1E:44:EC:BA

OUI Code

Unique Identifier

Comprised of 8 bytes delimited by :

Acts as a physical name tag for devices

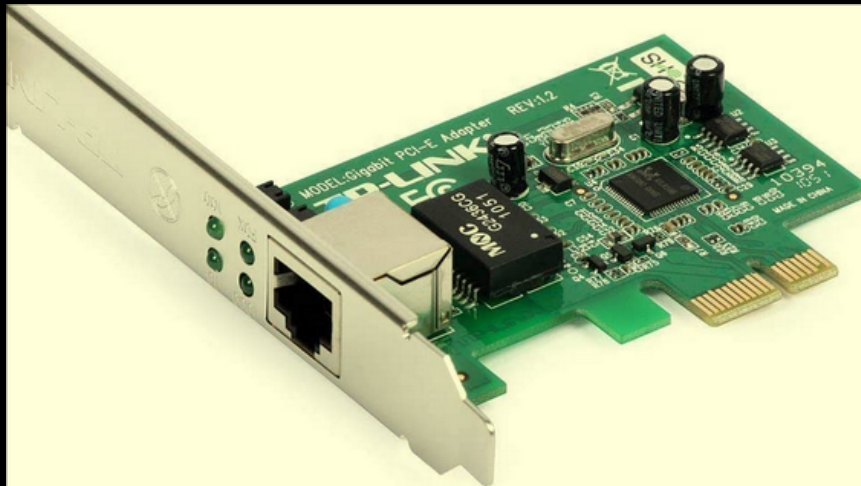
Network Interface

If you want to view your interfaces on your device:

- ip link show (linux)
- ipconfig (windows)

```
└─(13:40:17)─> ip link show ──(Sat,Mar21)─┐
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: enp2s0f0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc fq_codel state DOWN mode DEFAULT qlen 1000
   link/ether 8c:8c:aa:4e:f8:c2 brd ff:ff:ff:ff:ff:ff
   altname enx8c8caa4ef8c2
3: enp6s0f4u1u3: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc fq_codel state DOWN mode DEFAULT qlen 1000
   link/ether 00:50:b6:fd:13:40 brd ff:ff:ff:ff:ff:ff
   altname enx0050b6fd1340
4: wlp3s0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP mode DEFAULT qlen 1000
   link/ether e0:d4:64:57:2d:a3 brd ff:ff:ff:ff:ff:ff
   altname wlxe0d464572da3
5: tailscale0: <POINTOPOINT,MULTICAST,NOARP,UP,LOWER_UP> mtu 1280 qdisc fq_codel state UP mode DEFAULT qlen 1000
   link/none
7: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN mode DEFAULT qlen 1000
   link/ether 2e:fe:45:fa:29:8f brd ff:ff:ff:ff:ff:ff
```

Network Interface Controller (NIC)

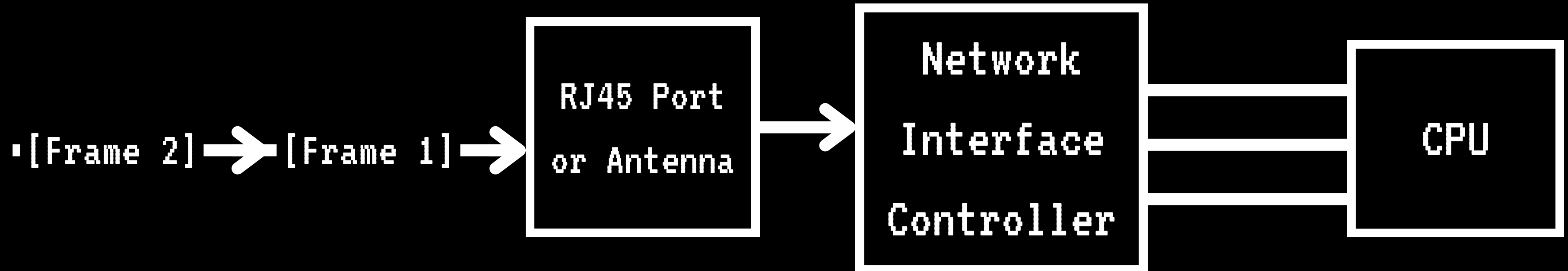


Physical card that controls network communication by providing network interfaces

Uses physical technology like:

- RJ-45 for ethernet
- Antenna for 802.11 wifi

Network Interface Controller (NIC)



Network Interface Modes

A network interface's traffic is determined by its mode:

- Managed Mode:
 - Accept only those with the interface's own MAC address
- Promiscuous Mode:
 - Accepts all packets regardless of dest MAC
- Monitor Mode:
 - Exclusive to wireless NIC
 - Captures raw 802.11 frames across all nearby channels

Network Architecture - Ad-hoc P2P

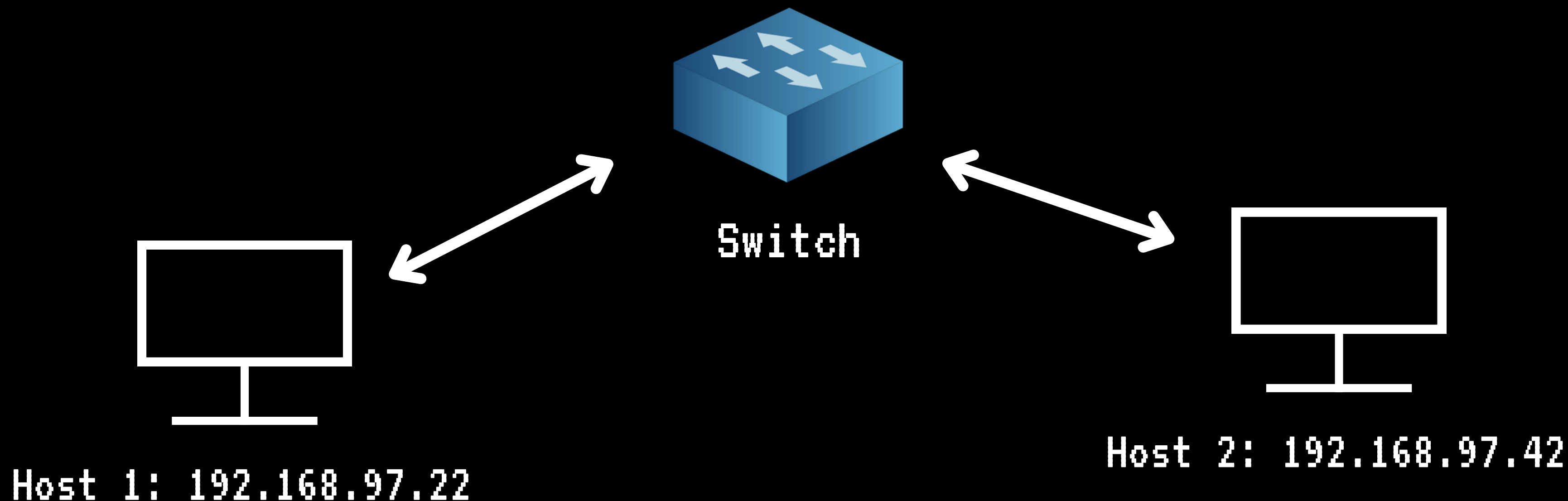
A network setup for directly communicating
with another device.



Network Architecture - LAN

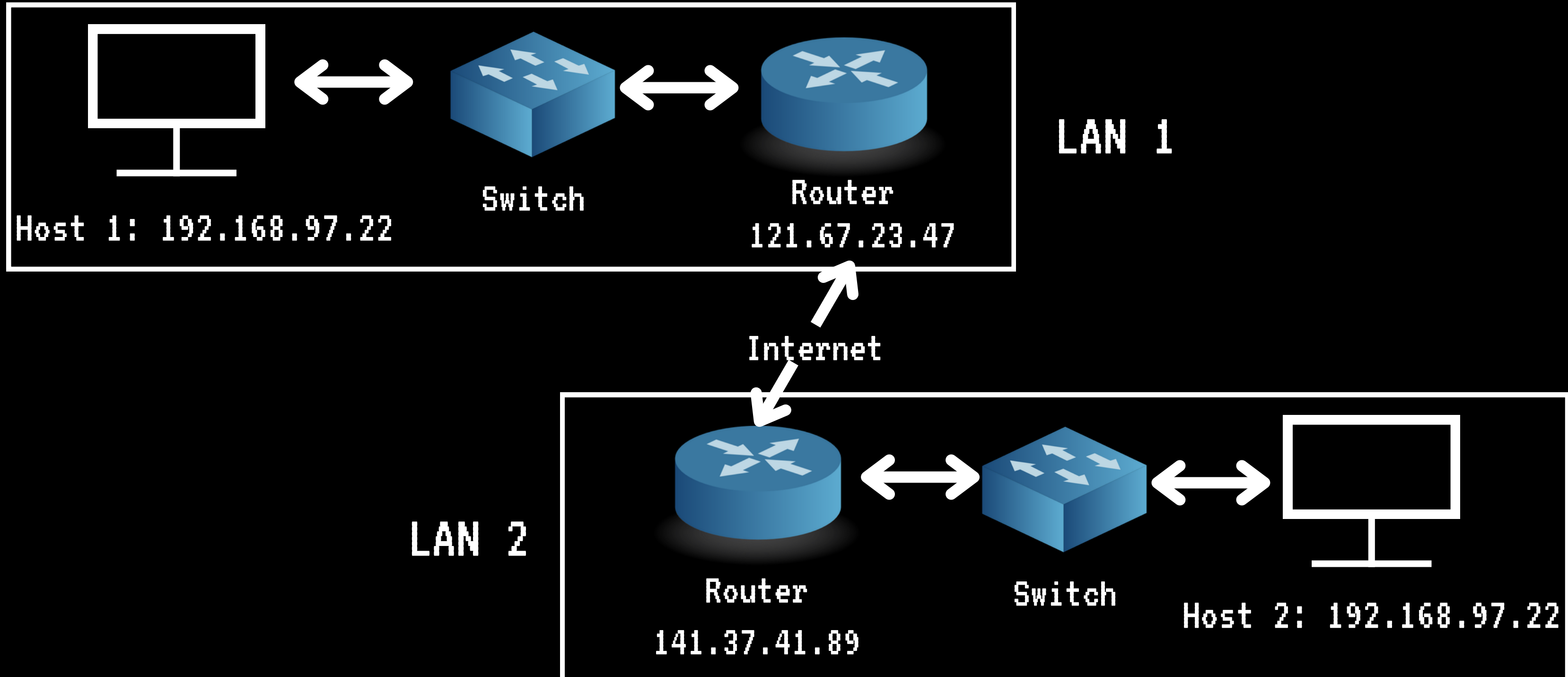
A local network setup for a home/small office.

Switch manages traffic between devices



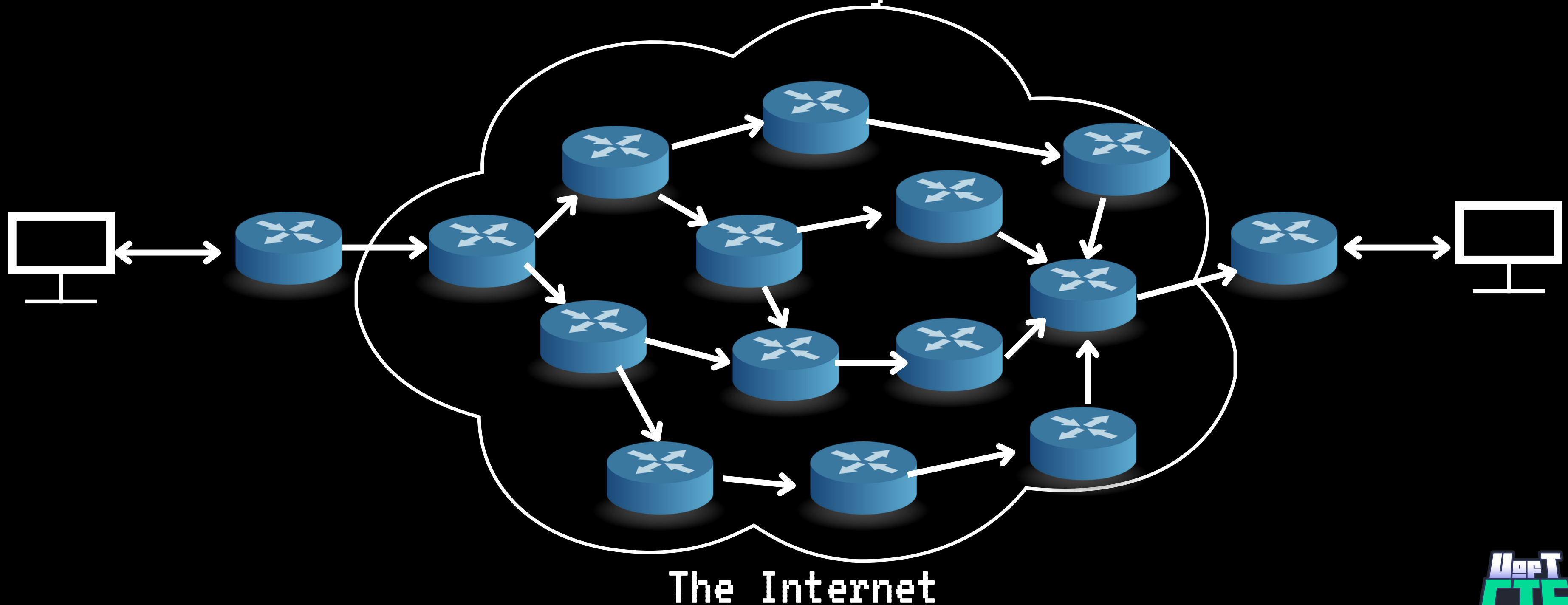
Network Architecture - WAN

A network setup for the internet. Devices communicate with internet gateways (routers) that forward requests to other routers



Sidenote - Internet

The internet is a collection of routers (some owned by ISP) that forward traffic long distances.



Internet Protocol (IP)

Protocol responsible for addressing of devices on a network.

Does this by providing an IP address

Two types:

- IPV4
- IPV6

IPv4 Address: (0-255).(0-255).(0-255).(0-255)

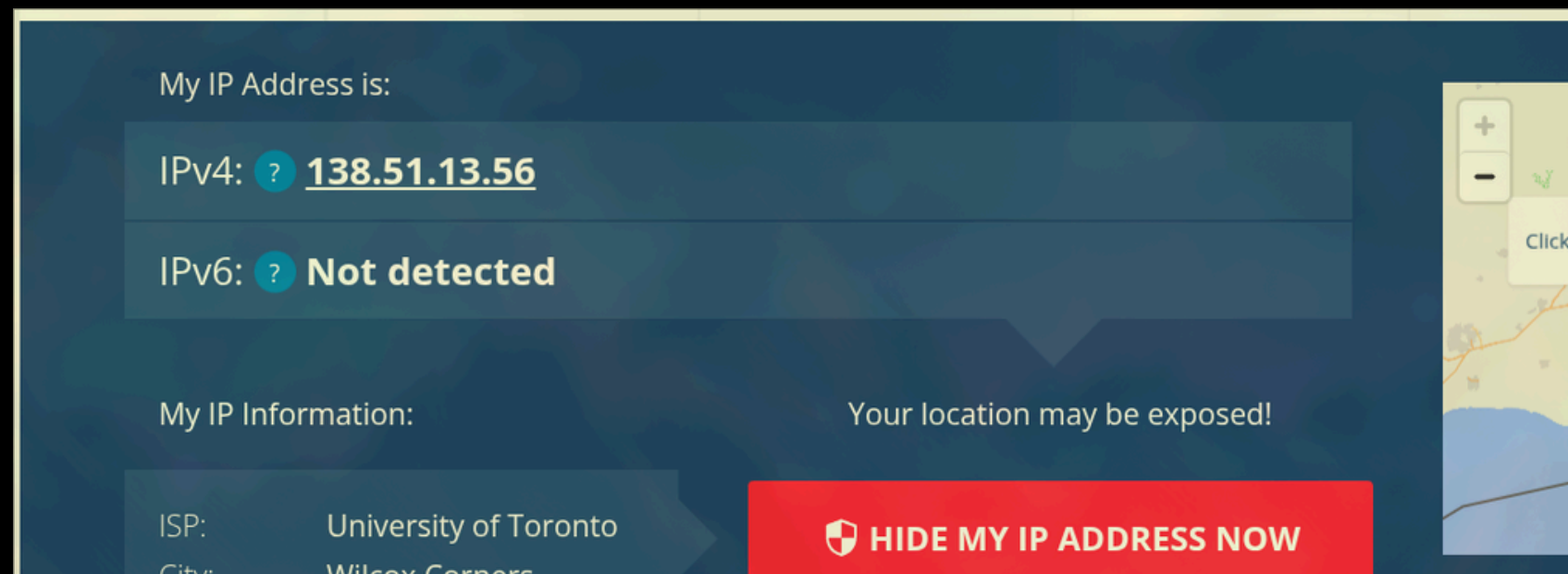
IPv6 Address: (0-ffff):(0-ffff):(0-ffff):(0-ffff):(0-ffff):(0-ffff):(0-ffff):(0-ffff)

Public/Private IPs

Every device has a:

- Private IP representing address for communication in LAN
- Public IP representing address of WAN-connected router

* Routers can be found geographically from IP



The screenshot shows a web application interface for displaying IP information. It includes a map on the right side. The text 'My IP Address is:' is followed by 'IPv4: ? 138.51.13.56' and 'IPv6: ? Not detected'. Below this, 'My IP Information:' is followed by 'ISP: University of Toronto' and 'City: Wilcox Corners'. A red button with a shield icon and the text 'HIDE MY IP ADDRESS NOW' is visible. A warning message 'Your location may be exposed!' is also present.

Public IP

```
8: cscotun0: <POINTOPOINT,MULTICAST,NOARP,
    link/none
    inet 100.98.141.208/19 brd 100.98.159.
        valid_lft forever preferred_lft for
    inet6 fe80::4cd9:4318:275e:7a66/126 sc
        valid_lft forever preferred_lft for
```

Private IP

Sidenote - CIDR

An IP address is broken down into two segments:

- Network ID (ID for the network)
- Host ID (ID for devices in the network)

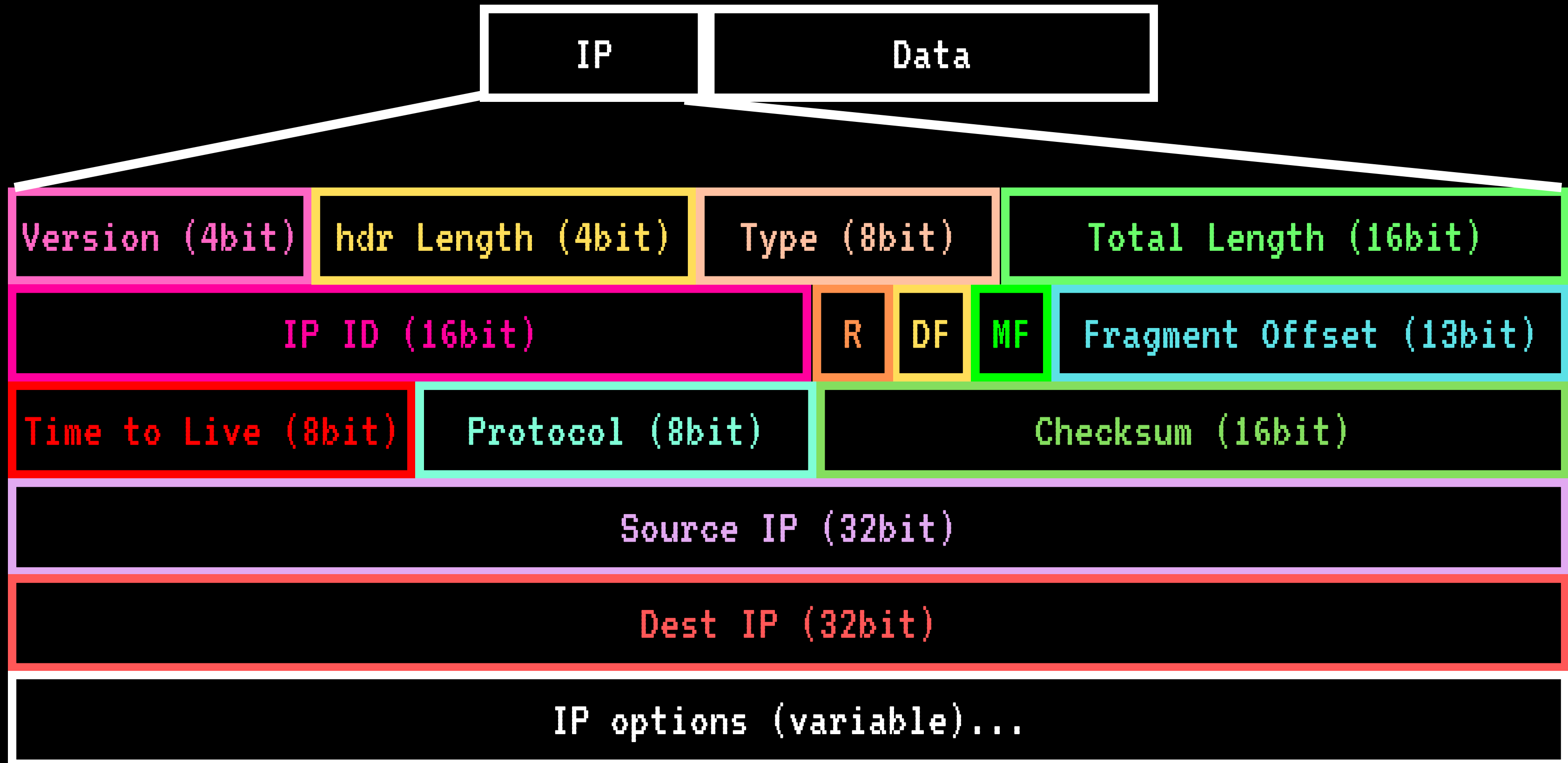


Class C:

- Public IP range: 192.0.0.0 - 223.255.255.0
- Private IP range: 192.168.0.0 - 192.168.255.255

* Look into class A,
class B in ur own time

IP Frame Header



IP Fragmentation

Messages are not sent in one big chunk.

Sometimes we have to break messages into smaller fragments to so that we dont surpass the **Maximum Transmission Unit**

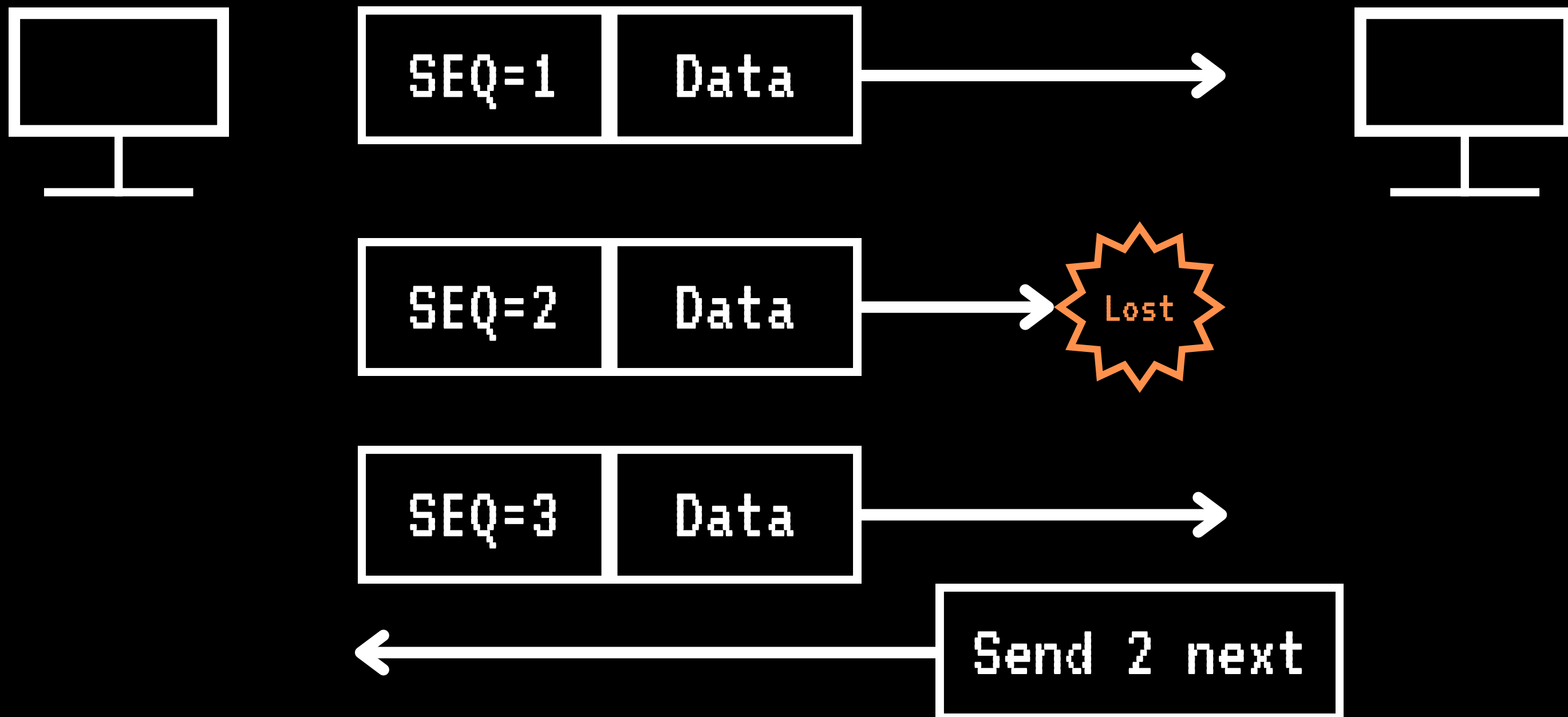


MTU: 2500 bytes



TCP

A protocol for sending/receiving messages that ensures
messages are kept in order



TCP Header

TCP

Data

Source Port (16bits)

Destination Port (16bits)

Sequence Number (32bits)

Acknowledgment Number (32bits)

Data offset (4bits)

Reserved (4bits)

CWR

ECE

URG

ACK

PSH

RST

SYN

FIN

Window (16bits)

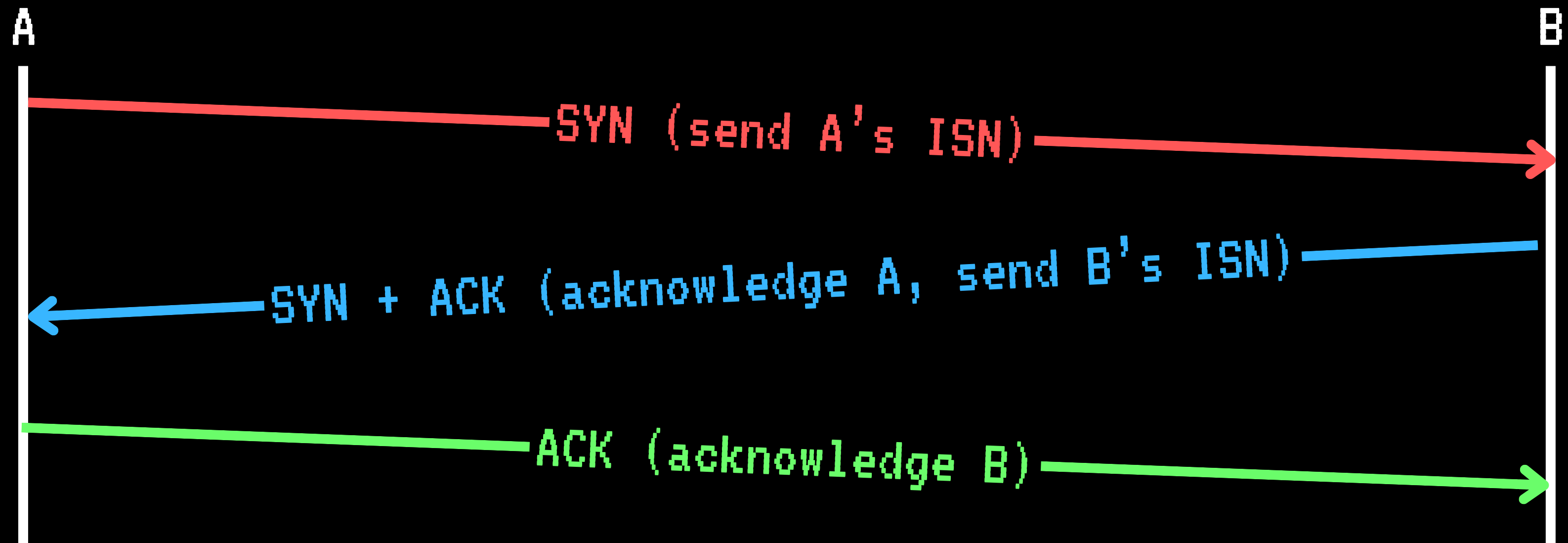
Checksum (16bits)

Urgent ptr (16bits)

Options (variable from data-offset)

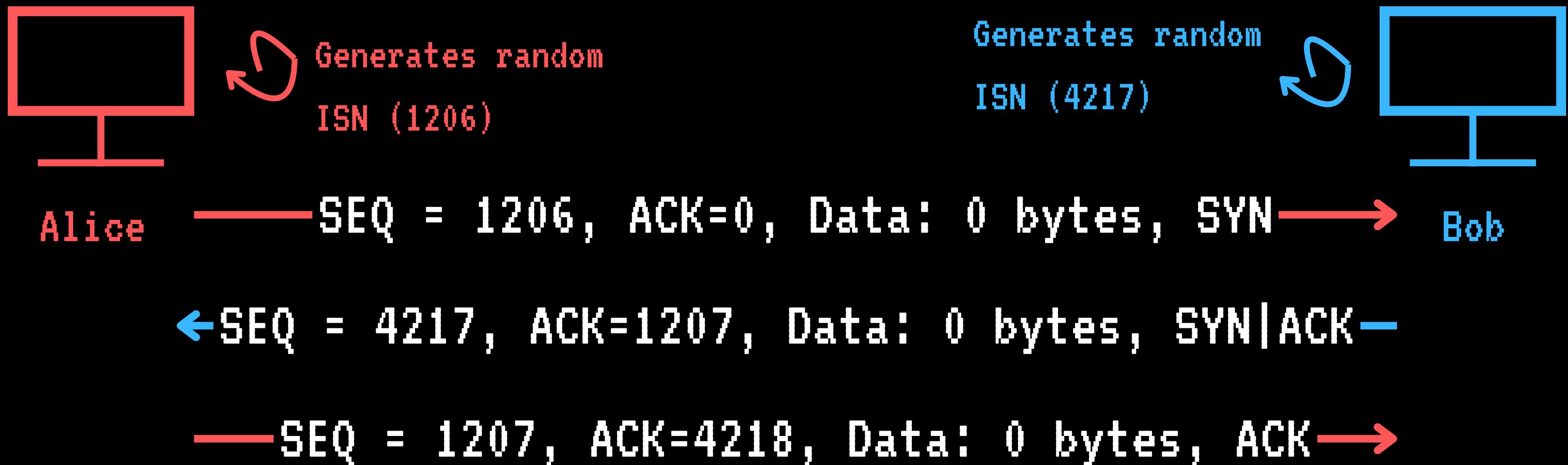
Three Way Handshake

Establish a session by sharing a initial sequence number (ISN) generated by both parties.



Three Way Handshake

Establish a session by sharing a initial sequence number (ISN) generated by both parties.

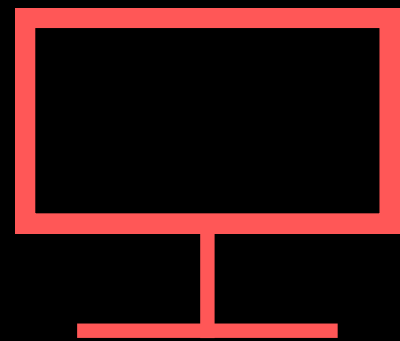


Done, can now begin communication!!

TCP Communication

Basic idea:

- Both devices store SEQ (bytes sent) and ACK (bytes received)
- On receive SEQ request, self ACK = given SEQ + payload size + 1
- On receive ACK response, self SEQ = given ACK



Alice

—SEQ = 1207, ACK=4218, Data: 200 bytes→

* initial SEQ from 3-way handshake

←SEQ = 4218, ACK=1408, Data: 20 bytes—

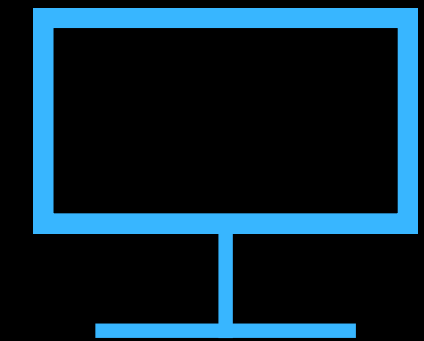
* initial SEQ from 3-way handshake

* ACK = SEQ + data size + 1 = 308

—SEQ = 1408, ACK=4239, Data: 10 bytes→

* SEQ = given ACK

←SEQ = 4239, ACK=1419, Data: 300 bytes—

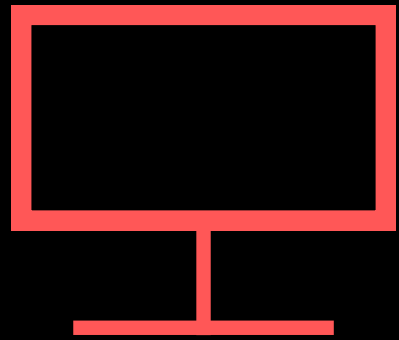


Bob

Delayed Acknowledgement

ACK does not always directly return, heres an optimization:

- Return an ACK after every second request
- If no second request after 200ms of first, return ACK



Alice

—SEQ = 1207, ACK=4218, Data: 200 bytes→

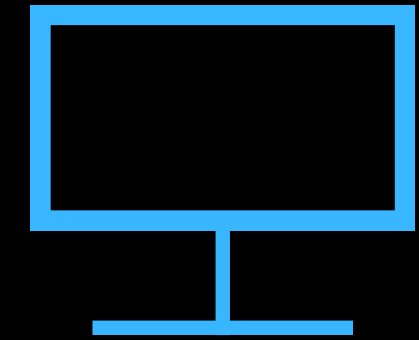
—SEQ = 1407, ACK=4218, Data: 10 bytes→

←SEQ = 4218, ACK=1418, Data: 0 bytes—

—SEQ = 1418, ACK=4219, Data: 10 bytes→

—SEQ = 1428, ACK=4219, Data: 2 bytes→

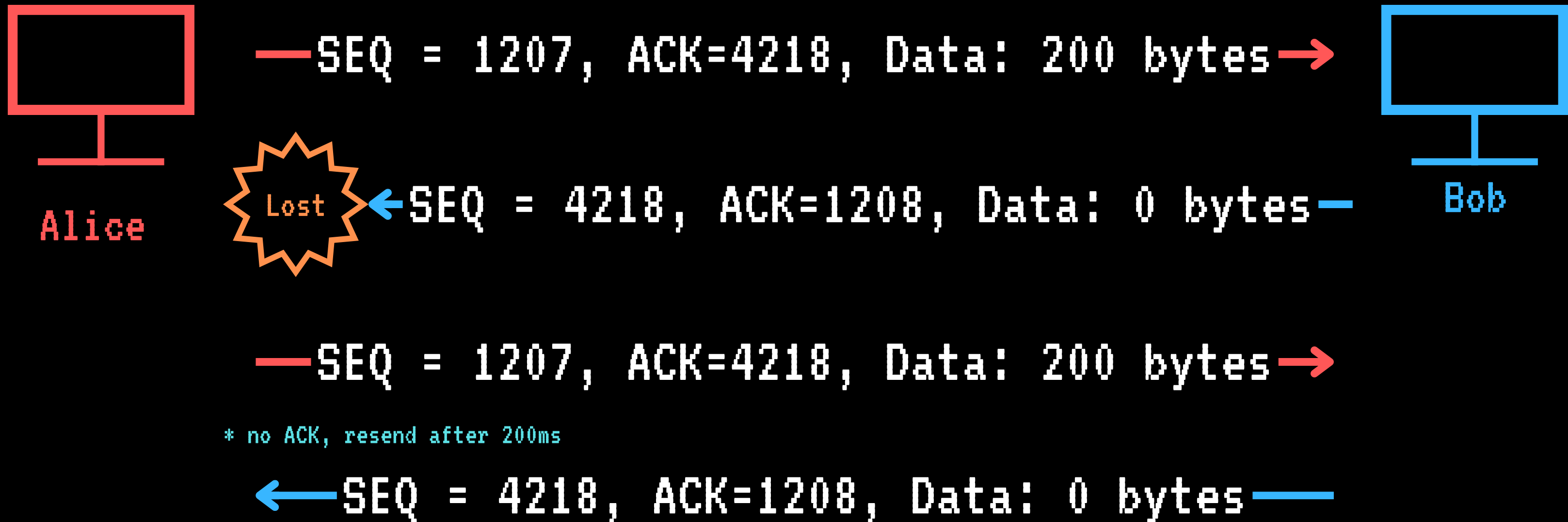
←SEQ = 4219, ACK=1441, Data: 10 bytes—



Bob

TCP Retransmission

If packets are lost during transmission (didn't receive a proper ACK), we must re-send these packets.



One Last Look at TCP Flags

The 8 flags in the TCP header:

CWR

ECE

URG

ACK

PSH

RST

SYN

FIN

- ~~CWR: window has been reduced~~
- ~~ECE: TCP is ECN capable~~
- ACK: acknowledgement field significant
- PSH: forces immediate ACK (no delayed acknowledgement)
- RST: resets TCP connection
- SYN: used for 3-way handshake initial seq number sync
- FIN: used to signal last packet from sender

Packet Captures (pcap)

These are captures of network traffic from one point of time to another

```
[ (23:07:46) ]> sudo tcpdump -i wlp3s0 -s 0 -X -n -vv -w output.cap
tcpdump: listening on wlp3s0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
^C77 packets captured
77 packets received by filter
0 packets dropped by kernel

[ /tmp/packets ]> ls
output.cap
```

Any program can generate these with libpcap.so or npcap

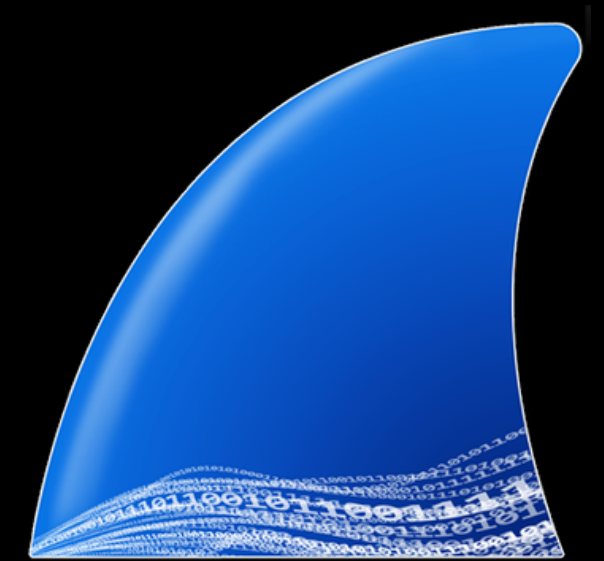
On linux you can use tcpdump or Wireshark

Wireshark

Network capture tool that automatically decodes packet captures

- Able to convert NIC into promiscuous or monitor mode

No.	Time	Source	Destination	Protocol	Length	Info
1191	2023-07-13 11:56:43.682651	192.168.101.1	192.168.101.36	DHCP	590	DHCP Offer - Transaction ID 0xc2a0fa93
1192	2023-07-13 11:56:43.704679	192.168.101.1	192.168.101.36	DHCP	590	DHCP ACK - Transaction ID 0xc2a0fa93
1192	2023-07-13 11:56:43.865487	192.168.101.36	192.168.101.1	TCP	78	49644 → 853 [SYN] Seq=0 Win=65535 Len=0 MSS=1220 SACK_PERM TSval=35069276
1192	2023-07-13 11:56:43.865569	192.168.101.1	192.168.101.36	TCP	60	853 → 49644 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
1192	2023-07-13 11:56:44.004298	192.168.101.36	192.168.101.1	DNS	89	Standard query 0x633f A connectivitycheck.gstatic.com
1192	2023-07-13 11:56:44.004383	192.168.101.36	192.168.101.1	DNS	74	Standard query 0x70c5 A www.google.com
1192	2023-07-13 11:56:44.012007	192.168.101.1	192.168.101.36	DNS	105	Standard query response 0x633f A connectivitycheck.gstatic.com A 172.217.
1192	2023-07-13 11:56:44.012103	192.168.101.1	192.168.101.36	DNS	90	Standard query response 0x70c5 A www.google.com A 172.217.26.228
1192	2023-07-13 11:56:44.033317	192.168.101.36	172.217.26.228	TCP	74	42498 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 SACK_PERM TSval=12805284
1192	2023-07-13 11:56:44.038110	172.217.26.228	192.168.101.36	TCP	74	443 → 42498 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1412 SACK_PERM TSv
1192	2023-07-13 11:56:44.065365	192.168.101.36	172.217.26.228	TCP	66	42498 → 443 [ACK] Seq=1 Ack=1 Win=65536 Len=0 TSval=1280528441 TSecr=2297
1192	2023-07-13 11:56:44.065449	192.168.101.36	172.217.26.228	TLSv1.3	633	Client Hello
1192	2023-07-13 11:56:44.065481	192.168.101.36	172.217.175.35	TCP	74	37918 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 SACK_PERM TSval=196528826
1192	2023-07-13 11:56:44.070188	172.217.175.35	192.168.101.36	TCP	74	80 → 37918 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1412 SACK_PERM TSv
1192	2023-07-13 11:56:44.070260	172.217.26.228	192.168.101.36	TCP	66	443 → 42498 [ACK] Seq=1 Ack=568 Win=66816 Len=0 TSval=2297825814 TSecr=17
1192	2023-07-13 11:56:44.071625	192.168.101.36	172.217.175.35	TCP	66	37918 → 80 [ACK] Seq=1 Ack=1 Win=65536 Len=0 TSval=1965288271 TSecr=38203
1192	2023-07-13 11:56:44.072105	192.168.101.36	172.217.175.35	HTTP	293	GET /generate_204 HTTP/1.1
1192	2023-07-13 11:56:44.075626	172.217.175.35	192.168.101.36	TCP	66	80 → 37918 [ACK] Seq=1 Ack=228 Win=66816 Len=0 TSval=3820324344 TSecr=196
1192	2023-07-13 11:56:44.075708	172.217.175.35	192.168.101.36	HTTP	212	HTTP/1.1 204 No Content
1192	2023-07-13 11:56:44.075738	172.217.175.35	192.168.101.36	TCP	66	80 → 37918 [FIN, ACK] Seq=147 Ack=228 Win=66816 Len=0 TSval=3820324345 TS
1192	2023-07-13 11:56:44.076872	192.168.101.36	172.217.175.35	TCP	66	37918 → 80 [ACK] Seq=228 Ack=147 Win=67072 Len=0 TSval=1965288276 TSecr=3



Colors packets based off protocol for easy analysis

Wireshark Streams

```
Wireshark · Follow TCP Stream (tcp.stream eq 520) · wireshark_pcapng_en0_20151115214627_FJxVdc

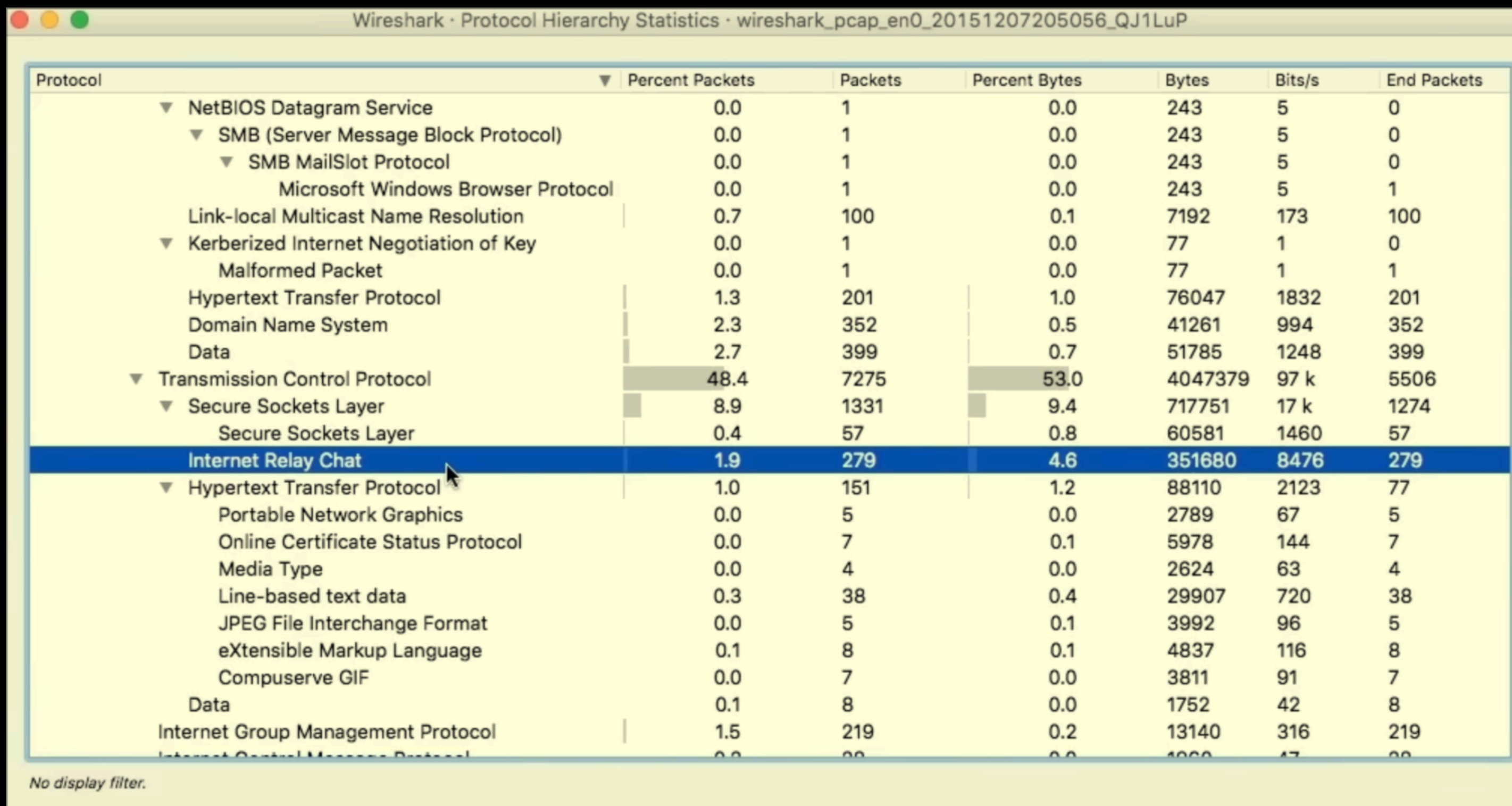
GET /ynewskatiecouric/lotus/b37e8d72-e5e4-475a-
abae-31942ac2d639_7RrRRilzWJaIHWgfjZZS67QM8ciA_qNunTcduVgcz7k6tY-JP0Wc23xu0dWltWTAUiHDsX-oKq0-
_3_0.m3u8?a=ynewskatiecouric&b=4950&ib=sapi&m=application%2fvnd.apple.mpegurl&mr=0&ns=c+i+ci+ci
+ps&ps=oxwrhyla&vid=fef9fe80-0c1b-370c-9929-814a6fd6f4a9&x=1448246859&s=1923041a8b7c96ab604de601ae2e6c
40 HTTP/1.1
Host: ci-cb577735f0-49103a02.http.atlas.cdn.yimg.com
Accept-Language: en-us
X-Playback-Session-Id: 01160DF6-4831-47DD-94D8-9BB5A423E936
Accept: */*
User-Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10_11_1) AppleWebKit/601.2.7 (KHTML, like Gecko)
Version/9.0.1 Safari/601.2.7
Referer: http://news.yahoo.com/
Accept-Encoding: gzip
Connection: keep-alive

HTTP/1.1 200 OK
Accept-Ranges: bytes
Date: Sun, 15 Nov 2015 16:22:21 GMT
Last-Modified: Sun, 15 Nov 2015 16:09:34 GMT
Server: ATS
Vary: Accept-Encoding
Content-Length: 6085
Connection: keep-alive
Content-Type: application/vnd.apple.mpegurl
Cache-Control: public, max-age=604800
Access-Control-Allow-Origin: *

#EXTM3U
#EXT-X-VERSION:3
#EXT-X-MEDIA-SEQUENCE:0
#EXT-X-ALLOW-CACHE:YES
#EXT-X-TARGETDURATION:9
```

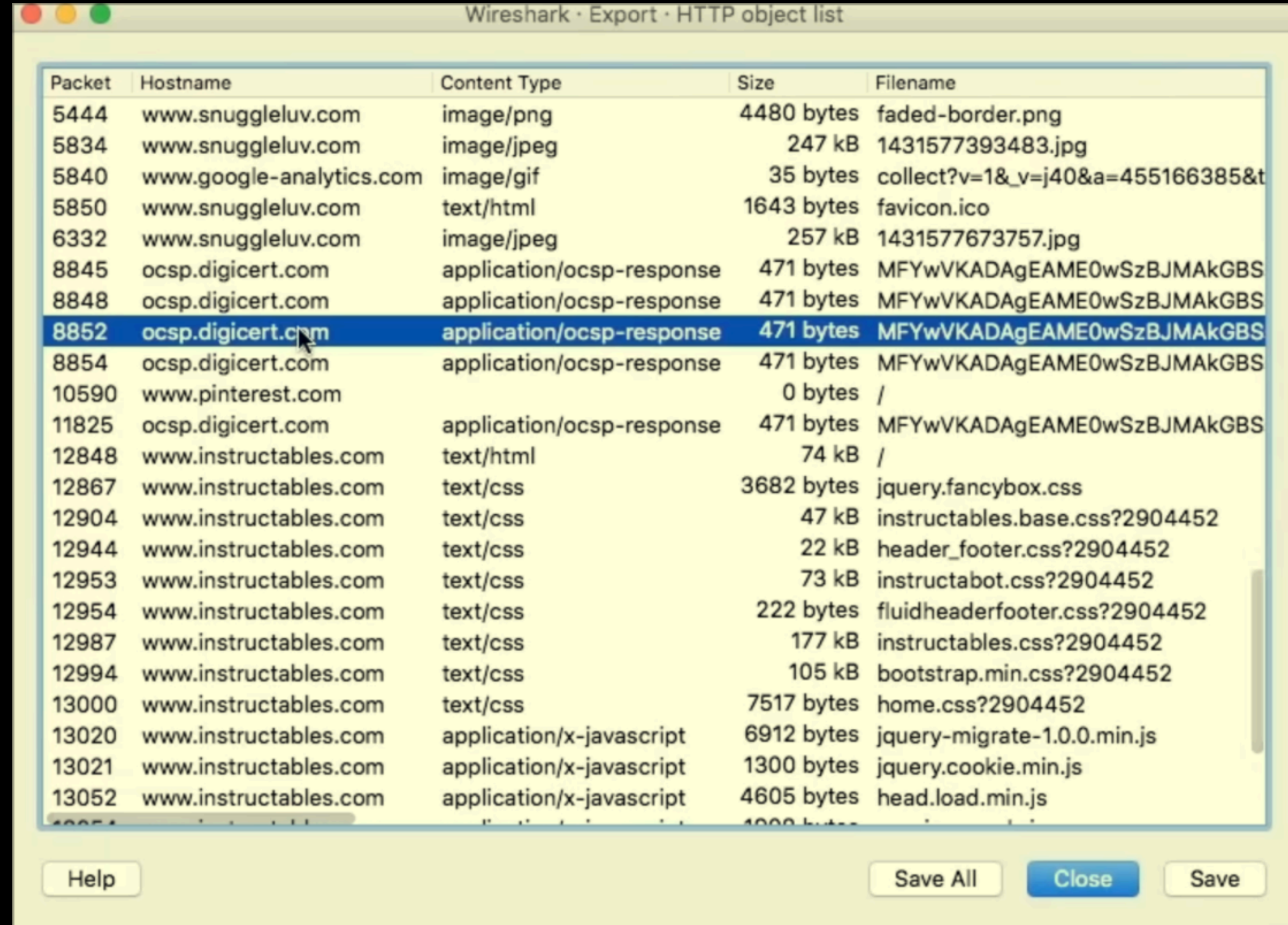
Right Click Packet > Follow > TCP Stream

Wireshark Protocol Hierarchy



Top Bar > Statistics > Protocol Hierarchy

Wireshark Export Objects

The image shows a screenshot of the 'Wireshark - Export - HTTP object list' dialog box. It features a table with five columns: Packet, Hostname, Content Type, Size, and Filename. The table lists various network objects, including images, HTML, and JavaScript files. The row for packet 8852 is highlighted in blue. At the bottom of the dialog, there are four buttons: 'Help', 'Save All', 'Close', and 'Save'.

Packet	Hostname	Content Type	Size	Filename
5444	www.snuggleluv.com	image/png	4480 bytes	faded-border.png
5834	www.snuggleluv.com	image/jpeg	247 kB	1431577393483.jpg
5840	www.google-analytics.com	image/gif	35 bytes	collect?v=1&_v=j40&a=455166385&t
5850	www.snuggleluv.com	text/html	1643 bytes	favicon.ico
6332	www.snuggleluv.com	image/jpeg	257 kB	1431577673757.jpg
8845	ocsp.digicert.com	application/ocsp-response	471 bytes	MFYwVKADAgEAME0wSzBJMAkGBS
8848	ocsp.digicert.com	application/ocsp-response	471 bytes	MFYwVKADAgEAME0wSzBJMAkGBS
8852	ocsp.digicert.com	application/ocsp-response	471 bytes	MFYwVKADAgEAME0wSzBJMAkGBS
8854	ocsp.digicert.com	application/ocsp-response	471 bytes	MFYwVKADAgEAME0wSzBJMAkGBS
10590	www.pinterest.com		0 bytes	/
11825	ocsp.digicert.com	application/ocsp-response	471 bytes	MFYwVKADAgEAME0wSzBJMAkGBS
12848	www.instructables.com	text/html	74 kB	/
12867	www.instructables.com	text/css	3682 bytes	jquery.fancybox.css
12904	www.instructables.com	text/css	47 kB	instructables.base.css?2904452
12944	www.instructables.com	text/css	22 kB	header_footer.css?2904452
12953	www.instructables.com	text/css	73 kB	instructabot.css?2904452
12954	www.instructables.com	text/css	222 bytes	fluidheaderfooter.css?2904452
12987	www.instructables.com	text/css	177 kB	instructables.css?2904452
12994	www.instructables.com	text/css	105 kB	bootstrap.min.css?2904452
13000	www.instructables.com	text/css	7517 bytes	home.css?2904452
13020	www.instructables.com	application/x-javascript	6912 bytes	jquery-migrate-1.0.0.min.js
13021	www.instructables.com	application/x-javascript	1300 bytes	jquery.cookie.min.js
13052	www.instructables.com	application/x-javascript	4605 bytes	head.load.min.js

Top Bar > File > Export Objects > HTTP

Wireshark Filters

Narrows down packet capture to filtered rules.

Example filters:

```
ip.addr == 10.0.0.145
```

Source IP is only 10.0.0.145

```
dns or http
```

Either DNS or HTTP traffic

```
tcp.srport == 443
```

Only source port 443 (usually tls)

Wireshark Conversations

Ethernet · 47 IPv4 · 81 IPv6 · 52 **TCP · 209** UDP · 402

Address A	Port A	Address B	Port B	Packets	Bytes	Packets A → B	Bytes A → B	Packets B → A	Bytes B
172.30.42.10	55870	172.30.42.13	445	10	600	5	330	5	
172.30.42.12	49259	172.30.42.13	445	12	720	6	396	6	
172.30.42.13	63818	10.0.0.8	7000	3	234	3	234	0	
172.30.42.13	63814	216.246.34.82	80	7	450	7	450	0	
172.30.42.13	63779	172.229.212.45	443	29	7313	19	4142	10	
172.30.42.13	63815	216.246.34.82	80	7	450	7	450	0	
172.30.42.13	63813	216.246.34.82	80	7	450	7	450	0	
172.30.42.13	63811	216.246.34.82	80	7	450	7	450	0	
172.30.42.13	63812	216.246.34.82	80	6	384	6	384	0	
172.30.42.13	59365	17.110.228.216	5223	12	1862	6	635	6	
172.30.42.13	63819	17.167.150.50	443	44	9375	23	5443	21	
172.30.42.13	63820	128.39.65.230	6667	465	363 k	203	13 k	262	
172.30.42.13	63669	45.33.18.155	443	9	696	5	361	4	
172.30.42.13	63821	10.0.0.5	7000	7	546	7	546	0	
172.30.42.13	63822	216.246.34.82	80	39	16 k	25	2091	14	
172.30.42.13	63823	216.246.34.82	80	28	7346	21	1769	7	
172.30.42.13	63824	17.167.150.73	443	31	7898	17	3530	14	
172.30.42.13	63825	216.246.34.82	80	48	22 k	30	2780	18	
172.30.42.13	63826	216.246.34.82	80	22	2248	19	1616	3	
172.30.42.13	63827	216.246.34.82	80	22	2303	19	1616	3	

☒ Name resolution ☐ Limit to display filter Conversation Types ▾

Help Copy ▾ Follow Stream... Graph... Close

Top Bar > Statistics > Conversations



Lets analyze a pcap file

<https://ctfd.yoshixi.net/>

[challenges#Undutmaning%20Kaptan%20Strang-9](https://ctfd.yoshixi.net/challenges#Undutmaning%20Kaptan%20Strang-9)

Memory Forensics

Volatile Memory (RAM)

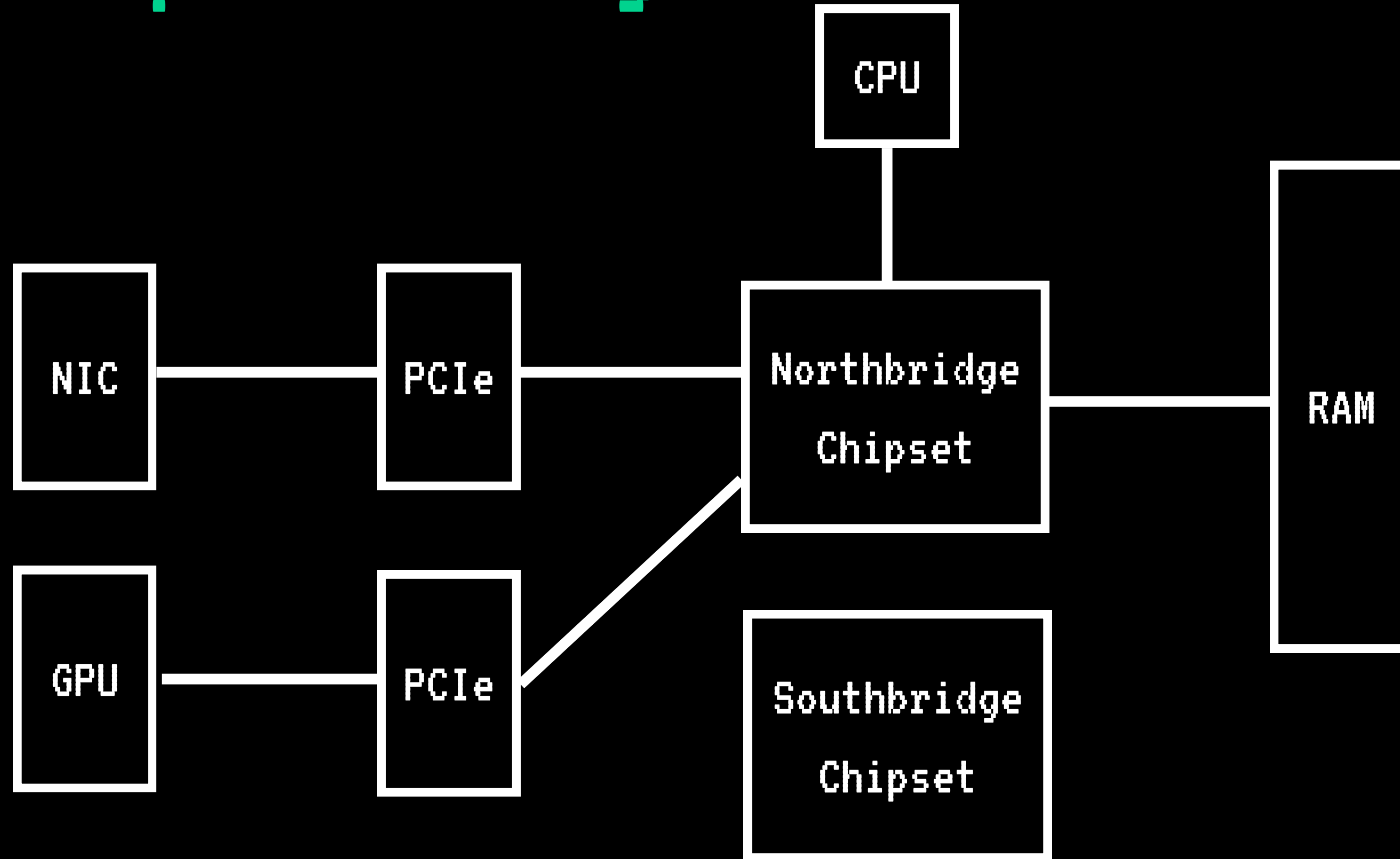
The main memory of the computer system.



Stored in DRAM chips that will have their state reset after power goes out (thats why it is volatile)



Computer Organization of Memory



Direct Memory Access: peripheral devices can directly access RAM

What does RAM Store?

- Running processes on the device
- Open file handles
- Active network streams
- Opened/Previously opened files
 - Unencrypted when reading
- Content in buffers (keystrokes)
- Loaded webpage contents (img, video)
- Sometimes entire OS (liveusb) (TailsOS)
- Kernel-use memory (Registry, pass hashes)

* mimikatz uses these in-memory password hashes to crack

 Shared Library

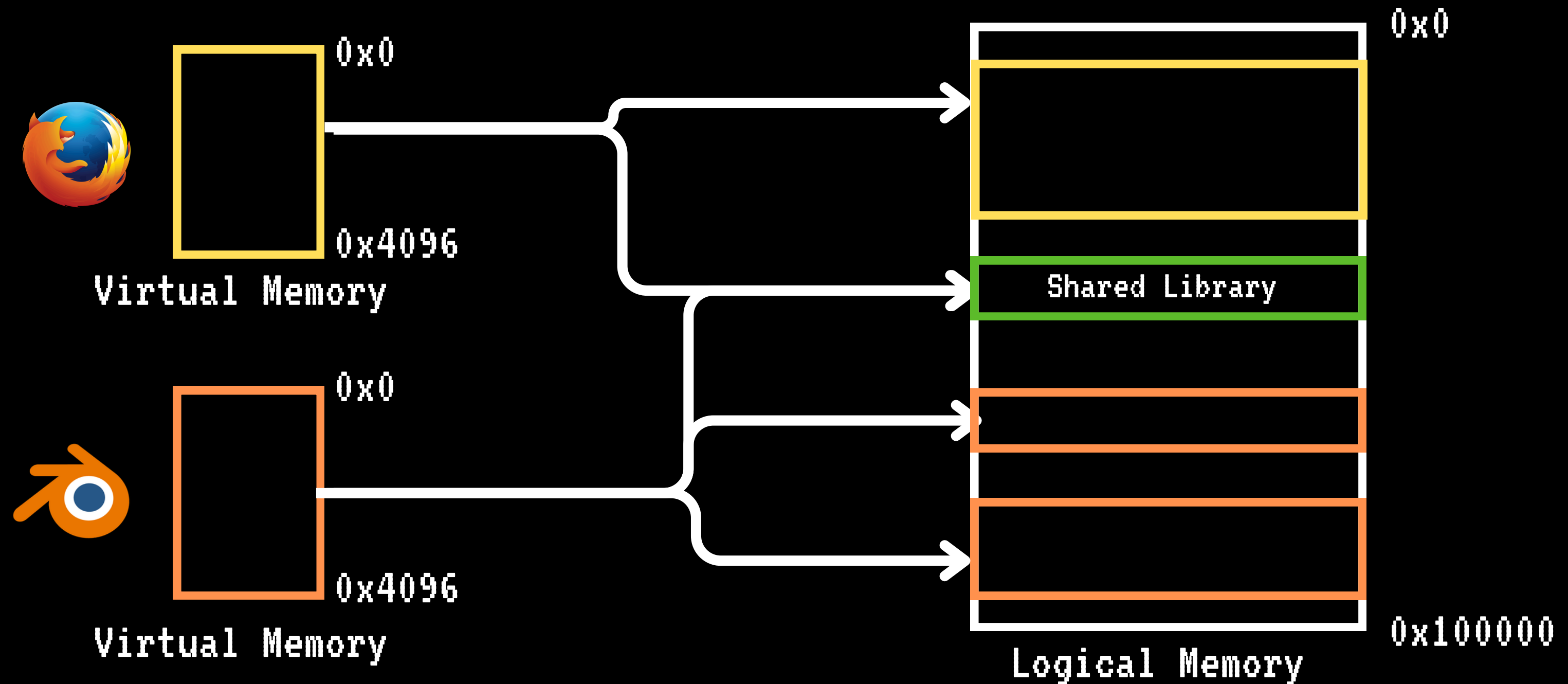
 Opened file

 Process

 Registry

Virtual Memory

A mapping from physical memory addresses to a virtual address space that the program can use instead



Volatility3

Simple tool used to analyze raw memory dumps

Requires setting up **symbol tables** which are json files to decode raw data into symbols

- Windows symbol tables are made public
- Linux symbols depend on distro, configs, might have to make your own



Volatility Installation

1. `git clone https://github.com/volatilityfoundation/volatility3.git`
2. `cd volatility3`
3. `pip install --user -e "[full]"`

If you want to setup a symlink:

1. `mv ../volatility3 /opt/volatility3`
2. `sudo ln -s /opt/volatility3/vol.py /bin/vol`

Using Volatility

```
vol -f <FILE> <PLUGIN_NAME> [<PLUGIN_OPTIONS>]
```

Nice plugins (windows dumps):

- `windows.pslist` : process list
- `windows.handles` : open file handles
- `windows.netscan` : network connections
- `windows.registry.printkey` : print registry key

Lets Analyze this Dump!

Logs & System Artifacts

Logs

Logs are persistent records of what has happened in:

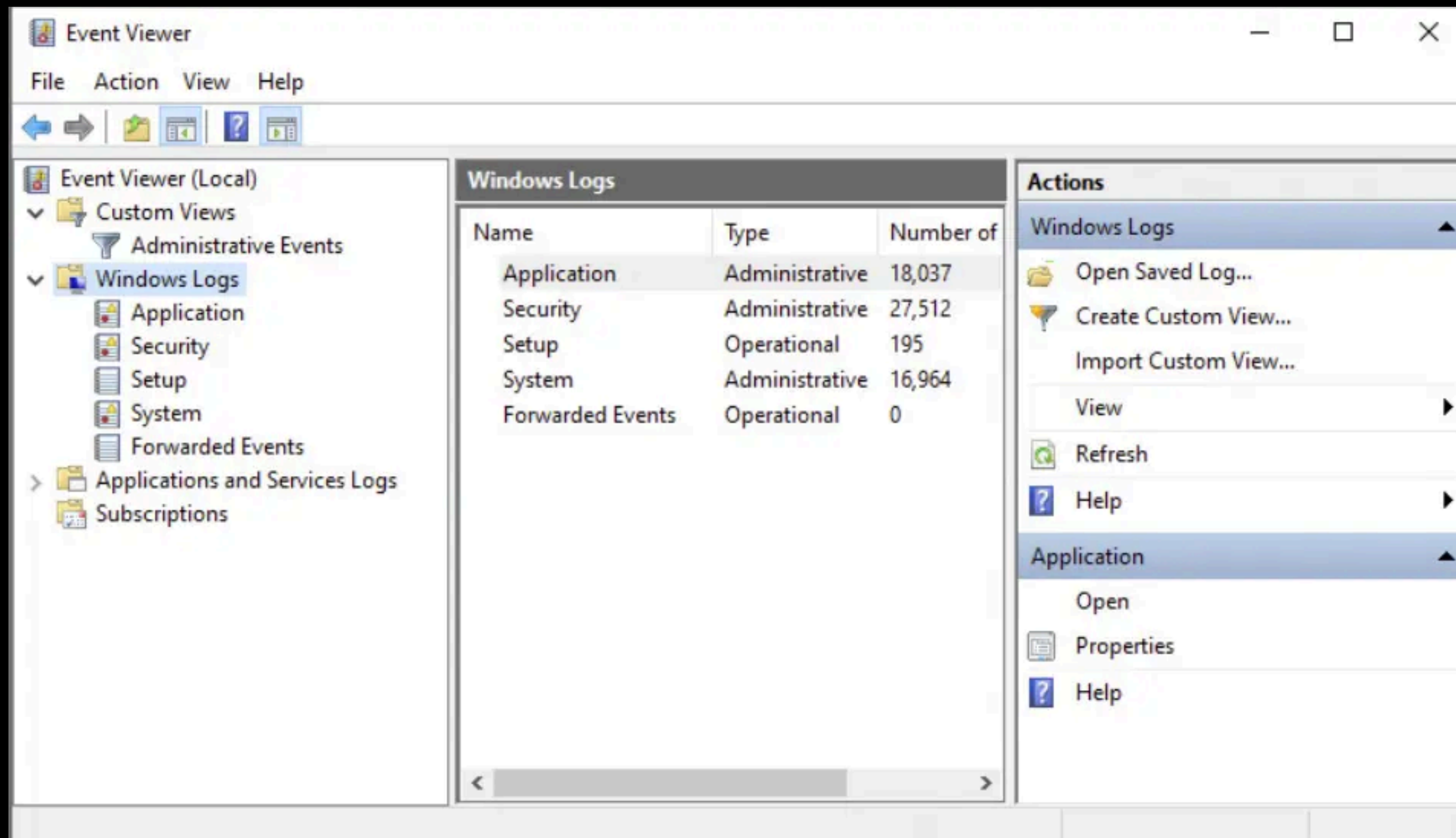
- A process
- The entire OS

Often stored in specific locations

- Windows: `C:\System32\winevt\Logs`
- Linux `/var/log`



Windows Event Viewer



GUI tool to view all log files and set filters for them

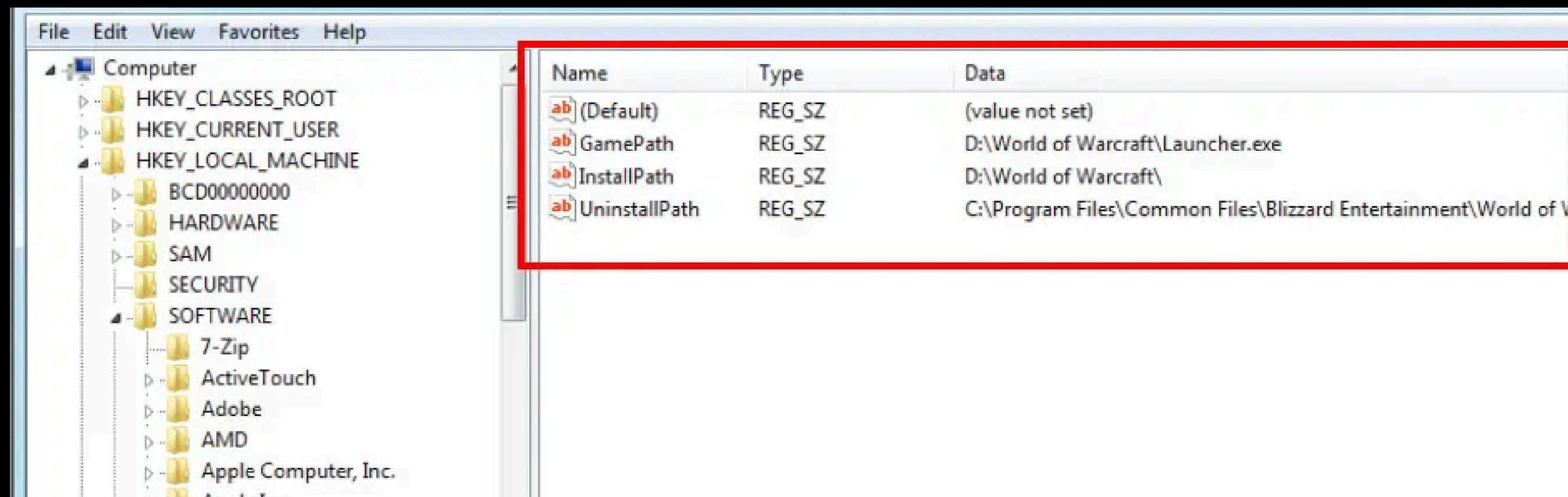
Windows Registry

Database storing configurations of system and programs. Comprised of **Hives**.



Hive: collection of keys, subkeys, values

Key: named entry corresponding to int, bool, string value



Windows Registry

Hives are stored at C:\System32\config

- SAM: user account info (username, passwords, login times, ...)
- SECURITY: local security policies
- SOFTWARE: computer software configs
- SYSTEM: system configs
- NTUSER.dat: user-specific configs
- DEFAULT: backup default system info

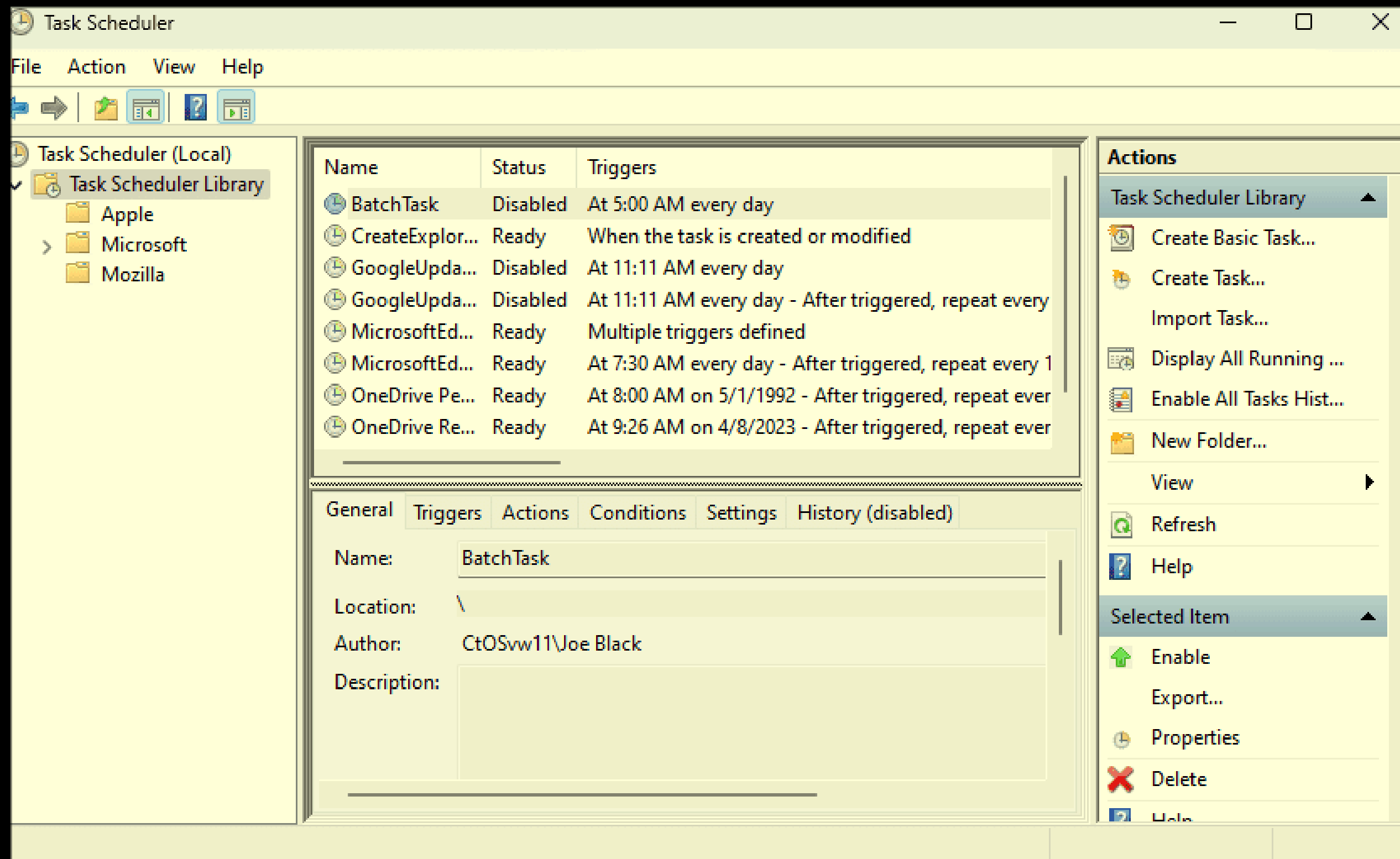
Windows Registry

* More: [https://
assets.tryhackme.com
/cheatsheets/
Windows%20Forensics%
20Cheatsheet.pdf](https://assets.tryhackme.com/cheatsheets/Windows%20Forensics%20Cheatsheet.pdf)

Some interesting hives and keys:

- C:\Windows\AppCompat\Programs\Amcache
 - Info about programs executed before
- SYSTEM\CurrentControlSet\Services\bam\UserSettings
 - Background process info
- SYSTEM\CurrentControlSet\Services\dam\UserSettings
 - Active process info
- Software\Microsoft\Windows\Shell\Bags
 - Details about all folders user accessed

Windows Scheduled Tasks



Processes may be scheduled to run in the future aswell

Resources

Giant wikis:

- <https://ctf.support/forensics/logs-and-system-artifacts/>
- <https://www.forensicxlab.com/blog/vmck>
- <https://hacktricks.wiki/en/generic-methodologies-and-resources/basic-forensic-methodology/partitions-file-systems-carving/file-data-carving-recovery-tools.html>
- <https://book.jorianwoltjer.com/forensics/wireshark>

Resources

Networking:

- <https://www.youtube.com/watch?v=n4muxtqLhN4>

Artifacts:

- [https://sigpwny.com/meetings/purple/2025-10-02/
Purple%20Team%2010_2_2025_%20Native%20Linux%20Forensics.pdf](https://sigpwny.com/meetings/purple/2025-10-02/Purple%20Team%2010_2_2025_%20Native%20Linux%20Forensics.pdf)
- [https://digitalforensics.ch/linux/practical-linux-forensics-
appendix-2021-07-17.pdf](https://digitalforensics.ch/linux/practical-linux-forensics-appendix-2021-07-17.pdf)

Filesystems:

- <https://www.youtube.com/watch?v=4KaF20XWfzg>

Resources

RAM Forensics:

- [https://repo.zenk-security.com/Forensic/The%20Art%20of%20Memory%20Forensics%20-%20Detecting%20Malware%20and%20Threats%20in%20Windows,%20Linux,%20and%20Mac%20Memory%20\(2014\).pdf](https://repo.zenk-security.com/Forensic/The%20Art%20of%20Memory%20Forensics%20-%20Detecting%20Malware%20and%20Threats%20in%20Windows,%20Linux,%20and%20Mac%20Memory%20(2014).pdf)
- <https://www.youtube.com/watch?v=fGP6VHxqkIM>